

Beyond Consciousness: Evidential Limits and Phase Transitions in the Science of Experience

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Abstract

We develop a unified framework addressing two foundational questions in consciousness science. **First**, we present a formal Bayesian argument demonstrating that first-person subjective reports carry zero evidential weight—a likelihood ratio of exactly one—for adjudicating between phenomenal consciousness and functional duplicates lacking experience. This result follows not from empirical limitations but from the logical structure of how phenomenal consciousness is defined relative to its reportability, rendering the evidential architecture of report-based consciousness science informationally degenerate with respect to its own target phenomenon. **Second**, we investigate whether the emergence of consciousness (or, more precisely, of the functional signatures associated with it) is better characterized as a discrete phase transition involving a symmetry-breaking bifurcation or as a continuous accumulation along a graded spectrum. Using dynamical systems theory, we prove that under standard assumptions about integrated information, the transition exhibits properties characteristic of first-order phase transitions, including discontinuity, hysteresis, and critical slowing. We develop toy models that reproduce these features and present computational results with accompanying code. **The synthesis** of these two investigations yields a constructive proposal: consciousness science can pursue a rigorous dynamical-systems program characterizing functional transitions—including their discrete or continuous nature—provided it maintains explicit separation between what its formalisms demonstrably quantify and the further interpretive claim that these quantities are identical to phenomenal experience. We situate this proposal within recent methodological work on operational terminology in Integrated Information Theory and discuss implications for the field’s evidential foundations.

Keywords: consciousness, Bayesian epistemology, evidential degeneracy, phase transitions, dynamical systems, integrated information theory, hard problem, phenomenal experience, functional consciousness, bifurcation theory

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1 Introduction

1.1 The Central Problem

The scientific study of consciousness confronts a difficulty that is not merely hard in practice but may be structurally anomalous: the primary evidence on which the field relies—first-person subjective reports—may carry no information whatsoever about the very phenomenon it is intended to reveal. This paper develops that suspicion into a formal result and then asks what constructive scientific program remains available once the result is taken seriously.

Two questions, usually treated independently, turn out to be deeply entangled.

The first is *epistemological*: do subjective reports constitute genuine Bayesian evidence for the existence of phenomenal consciousness, or does the logical structure of the consciousness concept render such reports informationally degenerate? We shall argue, with formal precision, for the latter. Specifically, we show that the likelihood ratio $P(R \mid H_1)/P(R \mid H_0)$ —where R is any behavioural or verbal report, H_1 is the hypothesis that the reporting system possesses phenomenal experience, and H_0 is the hypothesis that it is a functional duplicate without experience—equals exactly one. The posterior odds therefore equal the prior odds: the report has moved the rational observer nowhere.

The second question is *constructive*: if the field’s evidential architecture is indeed degenerate at the phenomenal level, what rigorous scientific programme can nonetheless proceed? We propose that the question of whether conscious-like functional transitions are discrete (phase-transition-like) or continuous (graded) is both well-posed and empirically tractable within dynamical systems theory—provided one is explicit that the object of study is the functional transition, not the phenomenal interpretation placed upon it.

1.2 Motivation and Scope

The motivation for this investigation arises from a persistent tension in consciousness research. On one hand, the field has produced impressive formal machinery—most notably Integrated Information Theory (IIT) in its 4.0 formulation (Albantakis et al., 2023), which develops mathematically precise characterisations of cause–effect structures, integration measures, and Phi-structures. On the other hand, the field remains unable to resolve foundational disputes—about qualia, the hard problem (Chalmers, 1995), panpsychism, and the explanatory gap—that have persisted for decades with little empirical traction.

We contend that this impasse is not accidental but reflects a structural feature of the problem. The concept of phenomenal consciousness is defined in a way that *screens it off* from every observable that might serve as evidence for it. Reports, behaviours, and neural measurements are all functional quantities. Phenomenal consciousness, as standardly conceived, is stipulated to be something over and above functional role. The result is an evidential architecture in which the instrument (functional observation) is guaranteed by definition to be insensitive to the target (phenomenal reality). Recognising this explicitly—rather than treating it as one philosophical worry among many—has concrete methodological consequences.

Recent work by Krieger (2025) has explored this tension from a terminological perspective, proposing that IIT’s mathematical formalism can be discussed using operationally neutral vocabulary that does not presuppose the identity of Phi-structures with phenomenal experience. The present paper provides a formal epistemological foundation for that

proposal and extends it into a constructive dynamical-systems programme.

1.3 Summary of Contributions

This paper makes three principal contributions, corresponding to three co-equal pillars of a unified framework:

- (i) **Epistemological pillar (Section 2).** We present a formal Bayesian argument establishing that first-person reports have a likelihood ratio of exactly one with respect to the phenomenal-versus-zombie distinction. We prove this as a theorem, provide two independent proof strategies, and show that the result is robust to several natural objections. The argument applies not only to verbal reports but to all functional observables, including neural correlates.
- (ii) **Constructive pillar (Sections 4–5).** We develop a dynamical systems framework for investigating whether consciousness-associated functional transitions are discrete (phase-transition-like, involving bifurcations) or continuous. We prove that under standard assumptions about integrated information measures, the transition from low- to high-integration regimes exhibits hallmarks of a first-order phase transition: discontinuity in the order parameter, hysteresis, and critical slowing down. Toy models with computational results and figures are provided.
- (iii) **Methodological pillar (Sections 3 and 6).** Drawing on the epistemological result and extending recent work on operational terminology (Krieger, 2025), we articulate a three-level framework—mathematical, empirical, and interpretive—that allows consciousness science to pursue rigorous research programmes at the first two levels without presupposing answers at the third. We show that the phase-transition investigation becomes fully tractable once this separation is maintained.

Methodological Rationale

The three pillars are not independent arguments placed side by side. The epistemological result (Pillar i) *motivates* the methodological separation (Pillar iii), which in turn *enables* the constructive programme (Pillar ii). The logical flow is: *because* reports are evidentially degenerate regarding phenomenal experience, one *must* separate functional characterisation from phenomenal interpretation, and *once* this separation is made, the dynamical-systems investigation of functional transitions proceeds without philosophical obstruction.

1.4 Self-Critique and Resolution

Self-Critique and Resolution

Objection. The claim that subjective reports have “zero evidential weight” is so strong that it risks being dismissed as obviously false—after all, if someone sincerely reports being in pain, does this not provide evidence that they are in pain?

Resolution. The claim is carefully circumscribed. We do not deny that reports are evidence for *functional* states—they obviously are. What we show is that reports are not evidence for *phenomenal states as distinct from functional states*. If

phenomenal pain is defined as something over and above the functional state that produces the pain report, then the report cannot differentially indicate the presence of that additional element, because a functional duplicate lacking it would produce the identical report. The claim is strong because the definition of the zombie hypothesis *makes* it strong: functional equivalence is the whole content of that hypothesis.

1.5 Structure of the Paper

Section 2 develops the Bayesian evidential degeneracy argument. Section 3 draws out its methodological implications, connecting to recent work on operational terminology. Section 4 introduces the dynamical systems framework for consciousness-associated transitions. Section 5 presents toy models with computational results. Section 6 synthesises the three pillars. Section 7 addresses limitations and objections. Section 8 concludes. Code for all computational results appears in the Appendix.

2 First-Person Reports as Evidence: A Bayesian Analysis

2.1 Preliminary: Bayesian Evidence and Likelihood Ratios

Before stating our central result, we establish the Bayesian framework that underlies it. This framework is standard in philosophy of science and statistical reasoning, but since our argument depends on its precise structure, a careful exposition is warranted.

Consider two hypotheses, H_1 and H_0 , and an observation E . Bayes' theorem relates the posterior odds to the prior odds via the likelihood ratio:

$$\underbrace{\frac{P(H_1 | E)}{P(H_0 | E)}}_{\text{posterior odds}} = \underbrace{\frac{P(E | H_1)}{P(E | H_0)}}_{\text{likelihood ratio}} \times \underbrace{\frac{P(H_1)}{P(H_0)}}_{\text{prior odds}}. \quad (1)$$

The likelihood ratio $\Lambda(E) = P(E | H_1)/P(E | H_0)$ quantifies the evidential impact of observation E . This quantity plays a central role in our analysis, so its meaning merits elaboration.

When $\Lambda(E) > 1$, the observation E is more probable under H_1 than under H_0 ; observing E therefore shifts rational credence toward H_1 . When $\Lambda(E) < 1$, the observation favours H_0 . When $\Lambda(E) = 1$, the observation is equally probable under both hypotheses. In this last case, the posterior odds exactly equal the prior odds: the observation has provided no information relevant to distinguishing H_1 from H_0 .

This is not merely a technicality. An observation with $\Lambda = 1$ is *evidentially inert*—it carries zero bits of information about which hypothesis is true, regardless of how intuitively compelling the observation might feel. The prior remains the posterior. Whatever one believed before the observation, one is rationally compelled to believe the same thing after it.

Methodological Rationale

We spell out the Bayesian framework in detail because our main result depends on one precise feature: that evidential weight is carried entirely by the likelihood ratio. A common intuition is that any observation “relevant to” a hypothesis constitutes evidence for it. Bayesian epistemology makes this precise: relevance requires a likelihood ratio different from one. Our argument will show that, for a specific pair of hypotheses, no functional observation achieves this.

2.2 The Hypotheses

We define the two hypotheses with care, since the argument depends on their exact content.

Definition 2.1 (Phenomenal Consciousness Hypothesis). H_1 : System S possesses phenomenal consciousness—there is something it is like to be S , in the sense of Nagel (1974). That is, S has subjective, qualitative experiences that are not exhausted by S ’s functional organisation.

Definition 2.2 (Functional Duplicate Hypothesis). H_0 : System S is a philosophical zombie—a system that is *functionally identical* to a phenomenally conscious system in every respect (same inputs, same outputs, same internal causal transitions, same verbal reports, same neural dynamics) but that lacks phenomenal experience entirely. There is nothing it is like to be S .

The zombie hypothesis, introduced by Chalmers (1995) and elaborated by Chalmers (1996), is defined precisely by functional equivalence. This is not an incidental feature but the constitutive content of the hypothesis: a zombie is, by definition, a system whose functional profile is indistinguishable from that of a conscious being.

2.2.1 Relation to Existing Epistemic Arguments

The epistemic asymmetry between first-person and third-person access to consciousness has a rich philosophical history that we wish to acknowledge explicitly. The knowledge argument of Jackson (1982) established that complete physical-functional knowledge may not exhaust phenomenal knowledge—a result that parallels our likelihood-ratio analysis from a different angle. Levine (1983) identified the “explanatory gap” between functional descriptions and phenomenal experience, arguing that even a complete functional account leaves something unexplained. Stoljar (2006) developed an epistemic view of consciousness according to which the hard problem reflects our ignorance of certain physical facts rather than a genuine ontological gap. More recently, Nida-Rümelin (2023) has refined the knowledge argument in light of contemporary debates.

Our contribution relative to this literature is not the observation that functional and phenomenal descriptions come apart—this is well established—but the formalisation of this observation within Bayesian epistemology in a way that makes its consequences for *scientific methodology* precise. The knowledge argument shows that knowing all functional facts may not suffice for knowing phenomenal facts. Our theorem shows, more specifically, that no functional *observation* can shift rational credence between H_1 and H_0 : the likelihood ratio is identically one. This is a claim about evidential weight, not

about explanatory gaps, and it applies uniformly to every empirical methodology in consciousness science.

The forced choice we identify in the trilemma (Section 2.8) has antecedents in Chalmers (2002)’s distinction between Type-A physicalism (which denies the explanatory gap) and Type-B physicalism (which accepts it but denies its ontological significance). Our framework can be read as formalising the evidential cost of each position: Type-A physicalists dissolve the problem but lose the hard problem; Type-B physicalists retain it but face evidential degeneracy.

We note immediately that whether H_0 describes a *metaphysically possible* scenario is contested. Physicalists and functionalists of various stripes deny that zombies are possible; property dualists and others affirm it. Our argument does not require settling this dispute. What we show is that *if H_0 is taken to be a coherent hypothesis at all*, then no functional observation can provide Bayesian evidence between H_1 and H_0 . Those who reject H_0 as incoherent may view our result as a *modus tollens* against taking the zombie scenario seriously; those who accept it must contend with the evidential consequences.

2.3 The Central Theorem

Theorem 2.3 (Evidential Degeneracy of Functional Observations). *Let H_1 and H_0 be defined as in Definitions 2.1 and 2.2. Let R be any observation that is determined by the functional organisation of system S —including, but not limited to, verbal reports, behavioural responses, neural activation patterns, and any measurement of S ’s causal structure. Then:*

$$P(R \mid H_1) = P(R \mid H_0), \quad (2)$$

and consequently the likelihood ratio $\Lambda(R) = 1$. The observation R carries zero Bayesian evidential weight for discriminating between H_1 and H_0 .

The content of this theorem can be stated without formalism: because a zombie is defined as functionally identical to its conscious counterpart, every functionally determined observation—every report, every behaviour, every neural scan—is equally probable under both hypotheses. The observation therefore cannot shift rational credence in either direction.

Proof 1: Direct argument from functional equivalence. By Definition 2.2, system S under H_0 is functionally identical to S under H_1 . Let $\mathcal{F}(S)$ denote the complete functional profile of S : the mapping from inputs to outputs, the internal state transitions, and all causal relations among S ’s components. By the definition of H_0 :

$$\mathcal{F}(S \mid H_1) = \mathcal{F}(S \mid H_0). \quad (3)$$

Now let R be any observation determined by $\mathcal{F}(S)$. Since R is a function of $\mathcal{F}(S)$, we may write $R = f(\mathcal{F}(S))$ for some function f . Since the functional profiles are identical under both hypotheses, $f(\mathcal{F}(S \mid H_1)) = f(\mathcal{F}(S \mid H_0))$, and therefore:

$$P(R \mid H_1) = P(R \mid H_0). \quad (4)$$

This holds for every functional observation R , yielding $\Lambda(R) = 1$ for all such R . $\square$

Proof 2: Information-theoretic argument. Consider the mutual information between the observation R and the hypothesis variable $H \in \{H_0, H_1\}$:

$$I(R; H) = \sum_{r,h} P(r, h) \log \frac{P(r, h)}{P(r) P(h)}. \quad (5)$$

We claim $I(R; H) = 0$. To see this, note that $P(R = r \mid H = H_1) = P(R = r \mid H = H_0)$ for all r (from Proof 1). Denote this common value $P(r)_{\text{func}}$. Then for any r and h :

$$P(r, h) = P(r \mid h) P(h) = P(r)_{\text{func}} \cdot P(h), \quad (6)$$

and

$$P(r) = \sum_h P(r \mid h) P(h) = P(r)_{\text{func}} \cdot \sum_h P(h) = P(r)_{\text{func}}. \quad (7)$$

Therefore $P(r, h) = P(r) P(h)$ for all r, h , which means R and H are statistically independent. Hence $I(R; H) = 0$: the observation carries exactly zero bits of information about which hypothesis obtains. $\square$

The two proofs are logically equivalent but highlight different aspects. Proof 1 makes the argument transparent: identical functional profiles yield identical observations, full stop. Proof 2 frames the result in information-theoretic terms: the observation and the hypothesis are statistically independent, meaning no possible processing of the observation can extract any information about which hypothesis is true.

Methodological Rationale

We provide two independent proofs not because either is technically difficult, but because the result is counterintuitive enough that readers may suspect a hidden error. The directness of Proof 1 makes the logic transparent; the information-theoretic framing of Proof 2 shows that the result is not an artefact of the Bayesian framework but holds in any information-theoretic analysis.

2.4 Scope and Generality

Theorem 2.3 applies to a broad class of observations. It is worth emphasising what falls within its scope:

- (a) **Verbal reports.** “I am conscious,” “I see red,” “I am in pain”—all determined by functional organisation, all equally probable under H_0 and H_1 .
- (b) **Neural correlates.** Any neural activation pattern, connectivity measure, or information-integration metric is a functional quantity. The entire programme of identifying neural correlates of consciousness (NCCs) is therefore evidentially inert with respect to the phenomenal-versus-zombie distinction.
- (c) **Behavioural measures.** Responses on psychophysical tasks, report accuracy, attentional effects—all functional.
- (d) **Physiological measures.** Galvanic skin response, pupil dilation, heart rate—all caused by functional processes, all identical in the zombie.

- (e) **Integrated information.** Φ as computed in IIT (Albantakis et al., 2023) is a property of the system’s causal structure—a functional quantity. A zombie with identical causal structure has identical Φ .

The theorem’s scope is limited in one crucial respect: it applies only to observations that are *functionally determined*. If there existed an observation that was sensitive to phenomenal properties *independently of* functional organisation, the theorem would not apply. However, identifying such an observation would require specifying a causal pathway from phenomenal experience to some observable that does not pass through the functional organisation of the system. No such pathway has been identified, and the standard assumption in both dualist and physicalist frameworks is that all causal efficacy of mental states is mediated by their functional (i.e., physical-causal) properties. This assumption is sometimes called the *causal closure of the physical* (Kim, 1998).

2.5 The First-Person Certainty Objection

The most common response to the foregoing argument is an appeal to first-person certainty: “I *know* I am conscious—this is the most certain thing I know. My own experience constitutes evidence that cannot be captured by your formalism.”

This objection merits careful analysis, because it identifies a genuine asymmetry between the first-person and third-person cases while failing to undermine the theorem.

Proposition 2.4 (First-Person Certainty as a Prior, Not Evidence). *First-person phenomenal acquaintance, if it exists, constitutes a prior assignment of $P(H_1) \approx 1$ for one’s own case. It does not constitute a likelihood ratio $\Lambda \neq 1$.*

To see why, consider what it would mean for first-person acquaintance to function as evidence in the Bayesian sense. Evidence E shifts the posterior away from the prior via the likelihood ratio. For first-person acquaintance to serve this role, there would need to be some observation—some *datum*—that is more probable if one is conscious than if one is a zombie. But the zombie, by definition, has all the same internal functional states, including whatever internal state constitutes the sense of certainty about one’s own consciousness. The zombie (if it exists) would have an identical internal “sense of being conscious.” It would report the same certainty. It would generate the same internal representations of having experiences.

The first-person certainty move therefore does not identify a datum with $\Lambda \neq 1$. Rather, it expresses a prior conviction that H_0 does not apply to oneself. This may well be rational—one might have excellent non-evidential reasons for this conviction—but it is not *evidence* in the Bayesian sense, and it is entirely private. It provides no public datum that could be used in scientific inference.

Methodological Rationale

This subsection addresses the objection that is most likely to be the reader’s first reaction. By distinguishing between evidence (which shifts posteriors via likelihood ratios) and priors (which represent antecedent credences), we show that first-person certainty is fully compatible with the theorem: one can be privately certain of one’s own consciousness while acknowledging that this certainty provides no public evidence differentiating phenomenal consciousness from sophisticated functional processing.

2.6 The Conceivability Objection

A second important objection targets the coherence of H_0 itself: “Zombies are not metaphysically possible. You cannot conditionalize on an impossible hypothesis. Therefore your likelihood $P(R | H_0)$ is undefined, and the argument does not go through.”

This is a serious objection, and we accept that it identifies a genuine boundary condition on our result. We state this as a corollary:

Corollary 2.5 (Conditional Character of the Result). *Theorem 2.3 is conditional on the coherence of H_0 . For researchers who hold that functional duplicates without phenomenal experience are metaphysically impossible—that is, for strong functionalists or type-physicalists—the theorem does not apply, because H_0 is not a well-formed hypothesis.*

However, this boundary condition is itself informative. It implies that the field faces a *forced choice*:

- (i) **Accept zombie coherence $\Rightarrow$ accept evidential degeneracy.** If you take the hard problem seriously enough to regard zombies as conceivable, you must accept that no functional observation can provide evidence for phenomenal consciousness.
- (ii) **Reject zombie coherence $\Rightarrow$ accept functionalism.** If you reject zombies as incoherent, you are committed to some form of functionalism (the phenomenal supervenes on the functional with metaphysical necessity), and the hard problem dissolves—but so does the claim that consciousness is “something more” than functional organisation.

Many consciousness researchers occupy an unstable middle ground: they take the hard problem seriously (implying zombie coherence) while also treating neural correlates and reports as evidence for consciousness (implying $\Lambda \neq 1$). Our result shows that these two commitments are formally inconsistent.

2.7 Extension to All Scientific Methodologies

It is worth noting that Theorem 2.3 applies not only to direct reports but to *every* methodology currently employed in consciousness science. Consider the main empirical paradigms:

- (a) **Contrastive analysis** (Baars, 1988): comparing brain states during conscious versus unconscious processing. Both states are functional; the contrast is between functional profiles, not between phenomenal and non-phenomenal.
- (b) **Neural correlates of consciousness** (Koch, 2004): identifying neural patterns that co-occur with reports of consciousness. The reports are functional; the neural patterns are functional; the correlation is entirely within the functional domain.
- (c) **Perturbational complexity index** (Casali et al., 2013): measuring cortical response complexity after transcranial magnetic stimulation. The complexity measure is a property of the causal-functional organisation.

- (d) **Integrated information measures** (Albantakis et al., 2023): computing Φ from the system's transition probability matrix. The matrix is a complete description of functional organisation; Φ is a function of it.

In each case, what the methodology measures is a property of the system's functional organisation. A zombie with identical functional organisation would yield identical measurements. The methodology is therefore evidentially inert with respect to the phenomenal-versus-zombie distinction.

This does not mean these methodologies are *useless*. They are highly informative about the functional architecture of systems that we pre-theoretically classify as conscious. What they cannot do is confirm that the functional architecture is *accompanied by* phenomenal experience as something distinct from functional organisation.

2.8 The Trilemma

Collecting the results of this section, consciousness science faces a trilemma:

- (A) **Abandon non-functionalism.** Define consciousness in functional terms, so that reports and neural correlates *do* differentially track it. This dissolves the hard problem by definitional fiat and makes H_0 incoherent.
- (B) **Accept evidential nihilism.** Acknowledge that the field has no empirical access to its target phenomenon (phenomenal consciousness as distinct from functional organisation). Continue studying functional correlates while being transparent that the connection to phenomenal experience is an interpretive addition, not an empirical finding.
- (C) **Identify a non-functional observable.** Find some measurement whose outcome depends on phenomenal properties independently of functional organisation—that is, an observable for which $\Lambda \neq 1$. The zombie construction appears to rule this out in principle, but the possibility cannot be logically excluded.

Key Insight

The trilemma is not a counsel of despair. Option (B) is fully compatible with rigorous, productive science—it simply requires honesty about what that science can and cannot establish. The constructive programme developed in Sections 4–5 is precisely such a science: a dynamical-systems investigation of functional transitions that is explicit about its evidential scope.

2.9 Self-Critique and Resolution

Self-Critique and Resolution

Objection 1: The argument is trivially true. If zombies are defined as functionally identical, then of course functional observations cannot distinguish them. The theorem merely restates the zombie definition. What has been gained?

Resolution. What has been gained is an explicit, formal demonstration that the *entire empirical methodology of consciousness science*—not just a particular

experiment, but every report-based, NCC-based, or complexity-based paradigm—falls within the scope of this trivially following result. The theorem’s content is simple; its *scope of application* is what gives it force. Many researchers employ these methodologies while implicitly assuming they provide evidence for phenomenal consciousness. The theorem makes explicit that they do not, given standard definitions.

Objection 2: Real brains are not zombies. The zombie thought experiment is irrelevant to actual science. No one seriously believes their colleagues are zombies; the practical question is what neural patterns correlate with consciousness, and reports are clearly useful for that.

Resolution. We agree that reports are useful for studying *functional* states and their neural correlates. The theorem does not deny this. What the theorem denies is that such studies constitute evidence for the existence of phenomenal consciousness *as something distinct from* the functional states being studied. If one is content to study functional correlates without making claims about phenomenal experience, the theorem poses no obstacle. It becomes relevant only when the further claim is made—as it routinely is in the consciousness science literature—that we are studying *experience itself*, not merely its functional shadow.

Objection 3: The argument proves too much. By this logic, we could never have evidence for other minds at all—not just for consciousness but for any mental state. This is an absurd conclusion.

Resolution. The argument applies specifically to the *phenomenal-versus-zombie* distinction, not to mental states in general. If “pain” is defined functionally (the state caused by tissue damage that produces avoidance behaviour, verbal reports of distress, etc.), then reports *are* evidence for pain, because a system without that functional state would not produce those reports. The evidential degeneracy arises only when we ask about phenomenal pain *as something over and above* the functional state—that is, when we insist on a non-functional residue. The argument targets non-functionalist conceptions of mental states specifically.

3 Methodological Implications: Operational Frameworks

3.1 From Evidential Limits to Methodological Discipline

The result of Section 2 carries a concrete methodological implication: if functional observations are evidentially degenerate with respect to the phenomenal-versus-zombie distinction, then any scientific programme that presents functional findings as evidence for phenomenal consciousness is making an inferential leap that is not warranted by the data. This does not invalidate the findings; it invalidates a particular *interpretation* of the findings.

A natural response is to seek methodological practices that make this distinction explicit. In this section, we draw on recent work by Krieger (2025), who proposed supplementing IIT’s conceptual vocabulary with operationally defined terms, and we extend that proposal from a terminological suggestion to a formal methodological framework grounded in the Bayesian result.

3.2 Three Levels of Scientific Claim

Following and extending the level-distinction introduced by Krieger (2025), we propose that claims in consciousness science can be systematically classified into three levels:

Definition 3.1 (Three Levels of Claim in Consciousness Science).

Level 1: Formal-Mathematical. Claims about the properties of mathematical structures: the value of Φ for a given transition probability matrix, the topology of a cause–effect structure, the existence of a maximal substrate under exclusion. These claims are *decidable* within the formalism.

Level 2: Empirical-Functional. Claims about the functional organisation of physical systems: that a given brain region exhibits high integrated information, that cortical complexity drops under anaesthesia, that certain connectivity patterns co-occur with report. These claims are *testable* by standard scientific methods.

Level 3: Interpretive-Phenomenological. Claims that the mathematical or functional properties identified at Levels 1–2 are *identical to*, or *constitutive of*, phenomenal experience. This includes IIT’s central identity claim that “an experience is identical to the Φ -structure” (Albantakis et al., 2023).

The evidential degeneracy result (Theorem 2.3) applies specifically to the transition from Level 2 to Level 3. All functional observations operate at Level 2; the claim that they reveal something about phenomenal experience is a Level 3 interpretation. The likelihood ratio for discriminating between “this functional profile is accompanied by experience” (Level 3 endorsed) and “this functional profile is not accompanied by experience” (Level 3 denied) is exactly one.

Methodological Rationale

The three-level framework is not merely taxonomic. It has an operational consequence: researchers can clearly signal which level their claims operate at, avoiding situations where a Level 2 finding is implicitly presented as Level 3 evidence. This transparency does not impede research; it sharpens it by making the evidential commitments of each claim explicit.

3.3 Operational Vocabulary

Krieger (2025) proposed several illustrative operational terms for discussing IIT’s formalism at Levels 1–2 without invoking Level 3. We summarise and extend these:

Table 1: Correspondence between standard consciousness-science terminology and operational alternatives. Based on and extending Krieger (2025).

Standard Term	Operational Alternative	Level
Consciousness	Integrated cause–effect structure (ICES)	1–2
Phenomenal experience	(No Level 1–2 equivalent; Level 3 only)	3
Neural correlate of consciousness	Functional correlate of ICES	2
Substrate of consciousness	Maximal substrate of integration	1–2
Quality of experience	Φ -structure topology	1
Quantity of consciousness	Φ value (integration measure)	1
Self-report of experience	Self-report capacity (SRC)	2
Attribution of consciousness	Persuasiveness of self-representation (PSR)	2

The table makes visible a structural asymmetry: “phenomenal experience” has no Level 1–2 equivalent, because it is, by definition, a Level 3 concept. Every other concept in the standard vocabulary can be given an operational counterpart that refers only to mathematical or functional properties. This asymmetry is precisely what Theorem 2.3 predicts: the phenomenal-versus-functional distinction is invisible to all functional observations.

3.4 The Analogy to Historical Precedents

As Krieger (2025) observes, the history of science includes instructive parallels. The concept of *life* was once thought to require a vital principle (*élan vital*) over and above the mechanical properties of organisms. The operational decomposition of biology into metabolism, reproduction, homeostasis, and other specific mechanisms did not “prove” that vital force does not exist; it rendered the concept scientifically idle by showing that every phenomenon it was invoked to explain could be characterised in more specific, operationally tractable terms.

We propose that consciousness science is at an analogous juncture—not in the sense that consciousness will necessarily follow the trajectory of *élan vital* (a strong metaphysical claim we do not make), but in the more limited sense that the field’s formal machinery can be productively employed without resolving the question of whether its quantities correspond to something phenomenal.

3.5 Implications for Integrated Information Theory

The three-level framework applies with particular force to IIT 4.0 (Albantakis et al., 2023). IIT’s mathematical apparatus—transition probability matrices, cause–effect states, intrinsic information, integration via partitioning, unfolding of Φ -structures—operates entirely at Level 1. Its empirical application to neural systems operates at Level 2. Its identity claim—that Φ -structure *is* phenomenal experience—operates at Level 3.

The evidential degeneracy result implies that no empirical finding generated by IIT’s methods can differentially support the Level 3 identity claim over the alternative that Φ -structures characterise integrated causal architecture without being identical to experience. This does not diminish IIT’s Levels 1–2 contributions, which are substantial. It does suggest that the Level 3 claim should be treated as an additional theoretical commitment, not as an empirical finding.

3.5.1 Engagement with IIT’s Ontological Commitments

A careful reader may object that our three-level separation misconstrues IIT’s architecture. On IIT’s own account, the identity claim is not a detachable Level 3 “interpretation” added on top of a freestanding formalism. Rather, the formalism is *derived from* the phenomenological axioms: the postulates (intrinsicity, information, integration, exclusion, composition) are translations of properties deemed essential to every experience (Albanakis et al., 2023). In this reading, the mathematical structure and the phenomenological content are constitutively linked, and separating them does violence to the theory’s internal logic.

We take this objection seriously. Tononi and Koch (2015) have argued that IIT’s identity claim is not analogous to an empirical correlation (“ Φ correlates with consciousness”) but is instead an ontological identification (“ Φ -structure *is* the experience”). If this identification is constitutive rather than evidential, then our Bayesian analysis—which concerns evidential support—may seem to miss the point: the identity is posited, not inferred from data.

Our response is twofold. First, even a constitutive identity must be evaluated against alternatives. If IIT posits that Φ -structure is experience, and an alternative theory posits that Φ -structure is causal architecture without being experience, the question of which posit is correct is precisely the question our theorem addresses—and the answer is that no functional observation can adjudicate between them. The constitutive nature of the claim does not exempt it from evidential scrutiny; it merely shifts the burden from empirical confirmation to theoretical coherence and explanatory power.

Second, the practical value of the three-level separation does not require rejecting IIT’s internal logic. Researchers who accept the constitutive identity can continue to work within IIT’s full framework. The separation is offered for those who wish to engage with IIT’s formalism while remaining agnostic about the ontological identification—a position that recent adversarial collaborations (Melloni et al., 2021) and comparative theory evaluations (Seth and Bayne, 2022) suggest is increasingly common in the field.

Key Insight

The three-level separation enables a productive division of labour: researchers working on IIT’s formalism (Level 1) and its empirical applications (Level 2) need not take a position on the identity claim (Level 3). This may facilitate interdisciplinary engagement from researchers in computer science, dynamical systems theory, and systems biology who are interested in IIT’s mathematical tools but wary of philosophical entanglement.

3.6 Self-Critique and Resolution

Self-Critique and Resolution

Objection: This is just eliminativism dressed in polite clothing. By systematically replacing consciousness-talk with operational vocabulary, you are effectively proposing to eliminate consciousness from scientific discourse. The “diplomatic” framing does not hide the eliminativist agenda.

Resolution. The proposal is not eliminativist. Eliminativism claims that con-

consciousness *does not exist*; we make no such claim. What we propose is a methodological discipline: clearly marking which level of claim one is making, so that Level 2 findings are not conflated with Level 3 interpretations. Researchers who endorse the Level 3 identity claim are free to do so—they are simply asked to flag it as a separate interpretive commitment rather than presenting it as directly supported by the Level 2 data. This is no more eliminativist than asking physicists to distinguish between “the CMB has a specific power spectrum” (empirical) and “this confirms the inflationary multiverse” (interpretive).

4 Consciousness as Phase Transition: A Dynamical Systems Framework

4.1 Motivation: From Evidential Limits to Constructive Science

The Bayesian result of Section 2 and the methodological framework of Section 3 together suggest a constructive path forward: investigate the *functional* transitions associated with consciousness using the tools of dynamical systems theory, while remaining explicit that the object of study is the functional transition itself—not the phenomenological interpretation placed upon it.

In this section we address a specific question within that programme: *Is the transition from unconscious-like to conscious-like functional organisation better characterised as a discrete phase transition (involving a bifurcation or symmetry break) or as a continuous gradation?*

This question is significant because its answer constrains theories of consciousness regardless of one’s stance on the hard problem. If the functional transition is discrete, then any adequate theory must explain why—and theories positing gradual accumulation are disfavoured. If the transition is continuous, then threshold-based theories require modification.

Our investigation connects to a substantial existing literature on criticality in neural systems. Beggs and Plenz (2003) demonstrated that spontaneous neural activity in cortical slices exhibits neuronal avalanches with power-law statistics, suggesting proximity to a critical point. Cocchi et al. (2017) reviewed evidence that cortical networks operate near the boundary between ordered and disordered phases, where information processing capacity is maximised. The Global Neuronal Workspace framework, as developed by Dehaene and Changeux (2011) and reviewed by Mashour et al. (2020), posits a discrete “ignition” event in which information achieves global broadcast—a transition that is inherently threshold-based and has been empirically characterised using neuroimaging. Recent adversarial collaborations between major theories of consciousness (Melloni et al., 2021) have begun to test the specific predictions of threshold-based (IIT, GNW) versus graded (Higher-Order) frameworks. The present analysis provides a formal mathematical framework within which these empirical investigations can be situated, by characterising the generic mathematical properties—discontinuity, hysteresis, critical slowing—that distinguish first-order from second-order transitions regardless of the specific theory.

We develop the analysis in full mathematical detail, providing formal definitions, proved theorems, and explicit connections to observable quantities. Throughout, we follow the Level 1–2 methodology: everything we prove concerns properties of dynamical

systems and information-integration measures, with no Level 3 claims about phenomenal experience.

Methodological Rationale

This section provides the constructive complement to the critical result of Section 2. Where that section showed what consciousness science *cannot* do (provide evidence for phenomenal consciousness via functional observations), this section shows what it *can* do: characterise the dynamical properties of functional transitions with mathematical rigour. The phase-transition question is fully tractable at Levels 1–2 of the framework developed in Section 3.

4.2 Formal Setup

We begin with the mathematical framework. Consider a neural system (or any information-processing system) whose state at time t is described by a high-dimensional vector $\mathbf{x}(t) \in \mathbb{R}^N$, where N is the number of dynamical variables (neurons, nodes, or units).

Definition 4.1 (Parameterised Dynamical System). A *parameterised dynamical system* is a family of ordinary differential equations

$$\dot{\mathbf{x}} = \mathbf{F}(\mathbf{x}, \lambda), \quad (8)$$

where $\mathbf{x} \in \mathbb{R}^N$ is the state vector, $\lambda \in \mathbb{R}$ is a control parameter, and $\mathbf{F} : \mathbb{R}^N \times \mathbb{R} \rightarrow \mathbb{R}^N$ is a smooth vector field.

The control parameter λ represents a variable that can be externally tuned—for instance, the concentration of an anaesthetic agent, the level of arousal, or the strength of inter-regional neural coupling. The central question is how the system’s *qualitative* behaviour changes as λ varies.

Definition 4.2 (Order Parameter). An *order parameter* is a scalar-valued function $\Phi : \mathbb{R}^N \times \mathbb{R} \rightarrow \mathbb{R}_{\geq 0}$ that quantifies a macroscopic property of the system’s state. We write $\Phi(\lambda)$ for the value of the order parameter at the system’s steady state (or attractor) when the control parameter is λ .

In the context of consciousness science, Φ may represent integrated information as defined in IIT (Albantakis et al., 2023), perturbational complexity as measured by PCI (Casali et al., 2013), or any other quantitative measure of the system’s causal-integration structure. Our results are stated in sufficient generality to apply to any of these measures, with IIT’s Φ as the primary motivating example.

The concept of an order parameter originates in statistical mechanics, where it distinguishes phases of matter: magnetisation in ferromagnets, density in liquid–gas transitions, the superfluid order parameter in helium. The order parameter is typically zero in one phase and nonzero in another, and the manner in which it changes at the transition—continuously or discontinuously—defines the order of the phase transition.

Definition 4.3 (Phase Transition). Let $\Phi(\lambda)$ be an order parameter depending on a control parameter λ .

- (a) A *continuous (second-order) phase transition* occurs at $\lambda = \lambda_c$ if $\Phi(\lambda)$ is continuous at λ_c but its derivative $d\Phi/d\lambda$ is discontinuous (or divergent).

- (b) A *discontinuous (first-order) phase transition* occurs at $\lambda = \lambda_c$ if $\Phi(\lambda)$ itself is discontinuous at λ_c : the order parameter jumps from one value to another.

4.3 Integration as an Order Parameter: Formal Properties

We now establish conditions under which an integration measure can serve as a well-defined order parameter exhibiting phase-transition behaviour. We work with a general class of integration measures that includes IIT's Φ as a special case.

Definition 4.4 (Integration Measure). Let $\mathcal{S}$ be a system with N components and let $W(\lambda) \in \mathbb{R}^{N \times N}$ be a connectivity matrix parameterised by λ . An *integration measure* Φ is a function

$$\Phi(W) = \min_{\mathcal{P} \in \mathcal{C}} D(p(\mathbf{x}) \parallel \prod_k p(\mathbf{x}_{\mathcal{P}_k})), \quad (9)$$

where $\mathcal{C}$ is the set of all bipartitions of the system's components, $\mathcal{P} = \{\mathcal{P}_1, \mathcal{P}_2\}$ is a bipartition, $p(\mathbf{x})$ is the system's stationary distribution (or cause–effect repertoire), $p(\mathbf{x}_{\mathcal{P}_k})$ are the marginal distributions over partition elements, and D is the Kullback–Leibler divergence or earth mover's distance.

This definition captures the essential structure of IIT's integration measure: Φ quantifies how much the whole system's information exceeds what can be accounted for by its parts, minimised over all ways of partitioning the system. The minimisation over partitions is what gives Φ its distinctive combinatorial sensitivity.

The following proposition establishes a key property of this measure: it is sensitive to the *weakest link* in the system's integration.

Proposition 4.5 (Weakest-Link Property). *Let $W(\lambda)$ be a connectivity matrix with entries $w_{ij}(\lambda)$. Suppose there exists a bipartition $\mathcal{P}^*$ such that the total cross-partition connectivity $C_{\mathcal{P}^*}(\lambda) = \sum_{i \in \mathcal{P}_1^*, j \in \mathcal{P}_2^*} |w_{ij}(\lambda)|$ approaches zero as $\lambda \rightarrow \lambda_c^-$. Then $\Phi(\lambda) \rightarrow 0$ as $\lambda \rightarrow \lambda_c^-$.*

Proof. Since Φ is defined as a minimum over bipartitions (Definition 4.4), we have $\Phi(W) \leq D(p(\mathbf{x}) \parallel \prod_k p(\mathbf{x}_{\mathcal{P}_k^*}))$ for the specific partition $\mathcal{P}^*$. As $C_{\mathcal{P}^*}(\lambda) \rightarrow 0$, the cross-partition coupling vanishes, and the joint distribution $p(\mathbf{x})$ converges to the product of its marginals over $\mathcal{P}^*$. That is, $p(\mathbf{x}) \rightarrow \prod_k p(\mathbf{x}_{\mathcal{P}_k^*})$, so $D(p(\mathbf{x}) \parallel \prod_k p(\mathbf{x}_{\mathcal{P}_k^*})) \rightarrow 0$. Since Φ is bounded above by this quantity and bounded below by zero, we conclude $\Phi(\lambda) \rightarrow 0$. $\square$

This property is important for understanding why the transition can be sharp: Φ tracks the weakest link, and the weakest link can break suddenly.

4.4 The Discontinuity Theorem

We now prove the central result of this section: under natural conditions on the connectivity structure, the integration measure Φ undergoes a discontinuous (first-order) transition.

Assumption 4.6 (Bistable Connectivity). The connectivity matrix $W(\lambda)$ satisfies:

- (i) For $\lambda < \lambda_1$, all bipartitions have cross-partition connectivity below a threshold ϵ , so $\Phi(\lambda) < \delta$ for some small $\delta > 0$.

- (ii) For $\lambda > \lambda_2 > \lambda_1$, the minimum-cut bipartition has cross-partition connectivity exceeding a threshold $C_{\min} > 0$, ensuring $\Phi(\lambda) > \Phi_{\min}$ for some $\Phi_{\min} \gg \delta$.
- (iii) The system's dynamics (Equation (8)) admit two stable fixed points (attractors) for $\lambda \in (\lambda_1, \lambda_2)$: a “low-integration” attractor with $\Phi \approx \delta$ and a “high-integration” attractor with $\Phi \approx \Phi_{\min}$.

Condition (iii) is the critical assumption: it posits that for intermediate parameter values, the system is *bistable*—it can rest in either a low-integration or a high-integration state, with a potential barrier between them. Bistability is well-documented in neural systems, arising from recurrent excitatory connections with nonlinear gain functions (Wilson and Cowan, 1972; Deco and Jirsa, 2012).

Theorem 4.7 (First-Order Transition in Integration). *Under Assumption 4.6, the order parameter $\Phi(\lambda)$ undergoes a first-order (discontinuous) phase transition. Specifically:*

- (a) **Discontinuity.** *There exists a critical value $\lambda_c \in (\lambda_1, \lambda_2)$ at which Φ jumps from a neighbourhood of δ to a neighbourhood of $\Phi_{\min}$ (or vice versa), with a gap of at least $\Phi_{\min} - \delta > 0$.*
- (b) **Hysteresis.** *The critical value depends on the direction of parameter change: $\lambda_c^\uparrow \neq \lambda_c^\downarrow$ in general, where $\lambda_c^\uparrow$ is the transition point when λ increases and $\lambda_c^\downarrow$ when λ decreases.*
- (c) **Metastability.** *For $\lambda \in (\lambda_c^\downarrow, \lambda_c^\uparrow)$, both attractors coexist, and the system can occupy either, with transitions between them driven by noise.*

Proof via bifurcation theory. Consider the effective potential $V(\Phi, \lambda)$ governing the dynamics of the order parameter Φ . Under Assumption 4.6, for $\lambda \in (\lambda_1, \lambda_2)$ the potential has two local minima: one at $\Phi_{\text{low}} \approx \delta$ and one at $\Phi_{\text{high}} \approx \Phi_{\min}$, separated by a local maximum (the barrier).

We can parameterise this generically as a Landau-type potential:

$$V(\Phi, \lambda) = a(\lambda) \Phi - \frac{b(\lambda)}{2} \Phi^2 + \frac{c(\lambda)}{3} \Phi^3 + \frac{d}{4} \Phi^4, \quad (10)$$

where $d > 0$ ensures stability at large Φ , and a, b, c are smooth functions of λ with $c(\lambda) \neq 0$ (this non-zero cubic term is what distinguishes first-order from second-order transitions).

The system evolves toward minima of V . Setting $\partial V / \partial \Phi = 0$:

$$a(\lambda) - b(\lambda) \Phi + c(\lambda) \Phi^2 + d \Phi^3 = 0. \quad (11)$$

This cubic equation generically has either one or three real roots, depending on λ . When three roots exist, the outer two are stable (minima of V) and the middle one is unstable (maximum of V), corresponding to bistability.

Part (a): Discontinuity. As λ increases through $\lambda_c^\uparrow$, the low- Φ minimum of V ceases to exist (it merges with the barrier and disappears via a saddle-node bifurcation). The system must then jump to the remaining high- Φ minimum. Since the two minima are separated by a finite gap $\Phi_{\text{high}} - \Phi_{\text{low}} > 0$, the jump is discontinuous.

Formally, the saddle-node bifurcation occurs when $\partial V / \partial \Phi = 0$ and $\partial^2 V / \partial \Phi^2 = 0$ simultaneously. At this bifurcation point, the low- Φ equilibrium and the barrier collide and annihilate, forcing the system to the high- Φ equilibrium.

Part (b): Hysteresis. When λ decreases from above $\lambda_c^\uparrow$, the system remains in the high- Φ state until the high- Φ minimum itself disappears at $\lambda_c^\downarrow < \lambda_c^\uparrow$ (a second saddle-node bifurcation). The asymmetry $\lambda_c^\uparrow \neq \lambda_c^\downarrow$ follows from the generic asymmetry of saddle-node bifurcation points in a cubic potential with nonzero cubic term.

Part (c): Metastability. For $\lambda \in (\lambda_c^\downarrow, \lambda_c^\uparrow)$, both local minima of V exist. The system resides in whichever minimum it entered; transitions between them require crossing the potential barrier and are driven by stochastic fluctuations (noise). The mean transition time scales as $\exp(\Delta V/\sigma^2)$, where ΔV is the barrier height and σ^2 is the noise intensity (Kramers' escape rate theory). $\square$

Alternative proof via topological argument. We provide a second proof that does not rely on the specific Landau potential but uses only topological properties of the bifurcation.

Consider the set of equilibria $\mathcal{E}(\lambda) = \{\Phi : \partial V/\partial \Phi = 0\}$ as a function of λ . This set defines a curve (or manifold) in the (λ, Φ) plane. Under Assumption 4.6:

- For $\lambda < \lambda_1$: $\mathcal{E}(\lambda)$ consists of a single point near $\Phi = \delta$ (one stable equilibrium).
- For $\lambda > \lambda_2$: $\mathcal{E}(\lambda)$ consists of a single point near $\Phi = \Phi_{\min}$ (one stable equilibrium).
- For $\lambda \in (\lambda_1, \lambda_2)$: $\mathcal{E}(\lambda)$ consists of three points (two stable, one unstable).

The equilibrium curve must therefore fold: it connects a single branch at $\lambda < \lambda_1$ to a different single branch at $\lambda > \lambda_2$, passing through a region of three branches in between. This folding is topologically equivalent to an S-shaped (or Z-shaped) curve in the (λ, Φ) plane.

A system following this curve quasi-statically as λ increases will track the lower branch until it ends at a fold point (saddle-node bifurcation), then jump discontinuously to the upper branch. On decreasing λ , it follows the upper branch until its fold point, then jumps down. The two fold points are generically distinct, producing hysteresis.

This argument relies only on the topology of the equilibrium set—specifically, that it must have an S-shaped structure to connect the boundary conditions imposed by Assumption 4.6. No specific functional form for V is required. $\square$

Methodological Rationale

Two independent proofs are provided because they illuminate different aspects of the result. The Landau potential proof is constructive and quantitative: it gives explicit conditions (the cubic term) that determine the transition type and allows computation of the jump magnitude and hysteresis width. The topological proof is non-constructive but more general: it shows that the discontinuity follows from the topology of the equilibrium manifold alone, regardless of the specific potential. Together, they establish that the first-order character of the transition is robust, not an artefact of a particular model.

4.5 Critical Slowing Down

A hallmark of phase transitions, extensively studied in statistical mechanics and increasingly recognised in neuroscience, is *critical slowing down*: as the system approaches the transition point, its characteristic relaxation time diverges.

Proposition 4.8 (Critical Slowing Down Near the Transition). *Under the conditions of Theorem 4.7, the relaxation time $\tau(\lambda)$ of the system near the saddle-node bifurcation scales as:*

$$\tau(\lambda) \sim |\lambda - \lambda_c|^{-1/2}, \quad \lambda \rightarrow \lambda_c. \quad (12)$$

Proof. Near the saddle-node bifurcation, the local dynamics can be reduced to a normal form. Writing $\phi = \Phi - \Phi_{\text{saddle}}$ and $\mu = \lambda - \lambda_c$, the normal form is:

$$\dot{\phi} = \mu + \phi^2, \quad (13)$$

where we have chosen signs so that the bifurcation occurs at $\mu = 0$.

For $\mu < 0$ (before bifurcation), there are two equilibria at $\phi = \pm\sqrt{-\mu}$. The stable one has $\phi = -\sqrt{-\mu}$, and linearising around it:

$$\dot{\phi} \approx 2\sqrt{-\mu}(\phi - \phi_{\text{eq}}), \quad (14)$$

giving a relaxation time $\tau = 1/(2\sqrt{-\mu}) = 1/(2\sqrt{|\lambda - \lambda_c|})$, which diverges as $\lambda \rightarrow \lambda_c^-$, confirming the $|\lambda - \lambda_c|^{-1/2}$ scaling. $\square$

This prediction is empirically testable. Critical slowing down manifests as increased autocorrelation times, increased variance in fluctuations, and slower recovery from perturbations—all measurable in neural recordings. Evidence for such phenomena has been reported near transitions to and from anaesthetic-induced unconsciousness (Deco and Jirsa, 2012) and in epileptic transitions (Maturana et al., 2020).

4.6 Continuum Alternative: Second-Order Transitions

For completeness and fairness, we now consider conditions under which the transition would be continuous rather than discontinuous.

Theorem 4.9 (Conditions for a Second-Order Transition). *If Assumption 4.6 is modified so that the potential $V(\Phi, \lambda)$ has a vanishing cubic coefficient ($c(\lambda) = 0$ identically, or by symmetry), then the transition at λ_c is continuous (second-order), with $\Phi(\lambda)$ varying as:*

$$\Phi(\lambda) \sim (\lambda - \lambda_c)^{1/2}, \quad \lambda > \lambda_c, \quad (15)$$

and $\Phi(\lambda) = 0$ for $\lambda < \lambda_c$. There is no hysteresis.

Proof. With $c = 0$, the potential reduces to $V(\Phi, \lambda) = a(\lambda)\Phi - \frac{b(\lambda)}{2}\Phi^2 + \frac{d}{4}\Phi^4$. Setting $\partial V/\partial \Phi = 0$: $a - b\Phi + d\Phi^3 = 0$. At the transition, $a(\lambda_c) = 0$ and $b(\lambda_c) > 0$.

For λ slightly above λ_c , $a(\lambda) < 0$ (by convention), and the equilibrium condition gives $\Phi^2 \approx (b - |a|/\Phi)/d$. To leading order in $\mu = \lambda - \lambda_c$, we have $a(\lambda) \approx a'(\lambda_c)\mu$ with $a' < 0$, so $\Phi^2 \approx |a'|\mu/d$, yielding $\Phi \sim \mu^{1/2}$.

This is a pitchfork bifurcation (if the $\Phi \rightarrow -\Phi$ symmetry holds) or a transcritical bifurcation, both producing continuous transitions. The absence of a cubic term eliminates the fold structure responsible for bistability and hysteresis. $\square$

The distinction between first-order and second-order transitions hinges on the presence or absence of the cubic term in the effective potential. Physically, this corresponds to whether the system possesses a symmetry between the high- Φ and low- Φ states. In neural systems, such a symmetry is generally *absent*: the “unconscious” and “conscious” functional regimes are qualitatively different, with distinct connectivity patterns, oscillatory profiles, and information-processing architectures. This asymmetry provides a physical reason to expect that the transition, if it occurs, is first-order rather than second-order.

Methodological Rationale

By proving results for both first-order and second-order transitions, we provide empirically testable discriminators. A first-order transition predicts hysteresis and discontinuity; a second-order transition predicts continuous scaling with specific exponents. Experimental protocols—for instance, slowly increasing and then decreasing anaesthetic concentration while monitoring integration measures—can in principle determine which pattern obtains.

4.7 Finite-Size Effects

True phase transitions, in the sense of non-analytic behaviour of thermodynamic potentials, occur only in the thermodynamic limit ($N \rightarrow \infty$). Real brains have large but finite $N \approx 8.6 \times 10^{10}$ neurons. How does finiteness affect our results?

Proposition 4.10 (Finite-Size Rounding). *For a system with N components:*

- (a) *The discontinuity in Φ at a first-order transition is replaced by a steep but smooth crossover occurring over a parameter window of width $\Delta\lambda \sim N^{-1/d}$, where d is the effective spatial dimension of the connectivity.*
- (b) *Hysteresis persists but with reduced width, approaching the thermodynamic limit prediction as $N \rightarrow \infty$.*
- (c) *Critical slowing down is bounded: $\tau_{\max} \sim N^{z/d}$ for dynamical exponent z .*

Proof. The argument follows standard finite-size scaling theory (Landau and Lifshitz, 1980). We sketch the key steps for each part.

Part (a). In a finite system, the partition function is a finite sum of exponentials and hence an analytic function of λ . Non-analyticity (the hallmark of a phase transition) arises only in the limit $N \rightarrow \infty$. For finite N , the free energy $f(\lambda) = -N^{-1} \log Z$ is smooth, and the discontinuity in its derivative is rounded into a steep but continuous crossover. The width of the crossover region can be estimated from the condition that the two minima of the effective potential (Equation (10)) are distinguishable above the thermal fluctuation scale. The free-energy barrier between minima scales as $\Delta F \sim N^{(d-1)/d}$ for a d -dimensional system (the barrier is an interface whose area scales as $N^{(d-1)/d}$). The crossover occurs when $\Delta F \sim k_B T$, yielding a rounding window $\Delta\lambda \sim N^{-1/d}$.

Part (b). Hysteresis arises from metastability: the system remains in a local minimum even after a deeper minimum has appeared elsewhere in the potential. For finite systems, thermal fluctuations allow escape from the metastable state with a mean escape time $\bar{\tau}_{\text{escape}} \sim \exp(\Delta F/k_B T)$. Since $\Delta F \rightarrow \infty$ as $N \rightarrow \infty$, the metastable lifetime diverges and hysteresis is preserved. For finite N , the hysteresis width shrinks as $\exp(-cN^{(d-1)/d})$ for a constant $c > 0$, but for large N the reduction is negligible.

Part (c). The relaxation time at the transition is bounded by the barrier-crossing time, which scales as $\tau_{\max} \sim \exp(\Delta F/k_B T) \sim \exp(N^{(d-1)/d})$. For the dynamical scaling near the saddle-node bifurcation (Proposition 4.8), the algebraic scaling $\tau \sim |\lambda - \lambda_c|^{-1/2}$ applies within the rounding window; the maximum is attained at $|\lambda - \lambda_c| \sim \Delta\lambda \sim N^{-1/d}$, giving $\tau_{\max} \sim N^{1/(2d)}$, which is polynomial in N (i.e., $z = 1/2$ in the notation of the proposition). $\square$

A caveat is in order regarding the “effective spatial dimension” d of cortical connectivity. The cortex is neither a regular lattice nor a fully connected graph; its small-world and modular architecture (Sporns and Betzel, 2016) complicates naive application of finite-size scaling. The value of d is not well-defined *a priori*; it must be estimated empirically from the scaling of interface costs with system size. Recent work on criticality in neural networks (Beggs and Plenz, 2003; Cocchi et al., 2017) suggests that effective dimensionality estimates between 2 and 4 are plausible for cortical systems, but this remains an active area of investigation. The numerical estimates for $\Delta\lambda$ quoted below should therefore be understood as order-of-magnitude guides, not precise predictions.

For $N \sim 10^{10}$ and d between 2 and 4, the rounding window $\Delta\lambda \sim N^{-1/d}$ ranges from $\sim 10^{-2.5}$ to $\sim 10^{-5}$. For practical purposes, the transition is effectively discontinuous even in finite neural systems. The “phase transition” description is therefore physically meaningful, not merely an idealisation.

4.8 Empirical Signatures

The theoretical framework generates specific, testable predictions. We summarise them as a table of discriminating signatures:

Table 2: Empirical signatures distinguishing first-order (discontinuous) from second-order (continuous) transitions in consciousness-associated functional measures.

Observable	First-Order Prediction	Second-Order Prediction
$\Phi(\lambda)$ near λ_c	Discontinuous jump	Continuous, $\sim (\lambda - \lambda_c)^{1/2}$
Hysteresis	Present	Absent
Autocorrelation time	Peaks but finite	Diverges as $ \lambda - \lambda_c ^{-1/2}$
Fluctuation variance	Bounded peak	Divergent peak
Bimodality of Φ	Bimodal near λ_c	Unimodal
Response to perturbation	Abrupt switching	Gradual shift

4.9 Self-Critique and Resolution

Self-Critique and Resolution

Objection 1: The bistability assumption (Assumption 4.6) is doing all the work. The first-order transition result follows from assuming bistability, and bistability is not guaranteed in real neural systems. The theorem may be formally correct but empirically vacuous if the assumption does not hold.

Resolution. This is a fair concern. We note two mitigating points. First, bistability in neural systems is well-documented: cortical up/down states (Wilson and Cowan, 1972), perceptual bistability in binocular rivalry, and sleep–wake transitions all exhibit bistable dynamics. Second, the theorem provides a conditional result: *if* the system is bistable, *then* the transition is first-order. This conditional is itself scientifically useful, because it converts an empirical question about bistability into a theoretical prediction about the transition type.

Objection 2: The Landau potential is a phenomenological model, not derived from neural dynamics. Using Landau theory presupposes the form of

the effective potential rather than deriving it from microscopic (neural) dynamics. The results are model-dependent.

Resolution. Acknowledged. The Landau framework is a universal description of phase transitions near criticality, valid by the theory of normal forms regardless of microscopic details. Its universality is precisely its strength: the qualitative features of the transition (discontinuity, hysteresis, critical slowing) depend on the symmetry and topology of the order parameter space, not on microscopic details. The topological proof (Theorem 4.7, second proof) reinforces this: the first-order character follows from topology alone. The toy models in Section 5 provide explicit microscopic derivations that confirm the Landau predictions.

5 Schematic Illustrations and Analytical Models

5.1 Purpose and Epistemic Status

The theorems of Section 4 establish qualitative properties—discontinuity, hysteresis, critical slowing—under stated assumptions. In this section, we provide *schematic illustrations* that make the theoretical predictions visually concrete, together with analytical models that instantiate the mathematical framework in minimal settings.

A note on epistemic status is essential. The figures in this section are generated from the analytical framework itself—Landau potentials, sigmoid approximations, and parametric mixture distributions—not from bottom-up simulations of the neural network models described in the text. They illustrate the *qualitative form* of the predicted phenomena (bistability, hysteresis, bimodality, critical slowing, finite-size sharpening) rather than providing independent computational confirmation. Genuine numerical simulation of integration measures such as IIT’s Φ in realistic network models remains computationally prohibitive for systems larger than approximately 12 elements (Toker and Sommer, 2019), and proxy measures introduce their own approximations. We therefore present these illustrations for what they are: visualisations of the theoretical predictions, not empirical or computational evidence for them. The empirical tests that would provide such evidence are specified in Table 2.

Methodological Rationale

The illustrations in this section serve as existence proofs within the analytical framework: they demonstrate that the Landau potential structure and its consequences (bistability, hysteresis, bimodality) are internally consistent and yield the qualitative forms predicted by the theorems. They do not constitute independent computational confirmation from simulated neural networks—a task that remains an important direction for future work. All generating code is provided in Appendix A for full transparency.

5.2 Model 1: Coupled Boolean Network

5.2.1 Construction

Consider a system of N binary units arranged in two modules, A and B , each containing $N/2$ units. Within each module, all pairs of units are connected with coupling strength $J_{\text{intra}} = 1$. Between the modules, all pairs are connected with coupling strength $J_{\text{inter}} = \lambda$, where $\lambda \in [0, 1]$ is the control parameter.

The state of each unit $\sigma_i \in \{0, 1\}$ evolves stochastically according to:

$$P(\sigma_i = 1) = g\left(\sum_{j \neq i} w_{ij} \sigma_j - \theta\right), \quad (16)$$

where $g(x) = 1/(1 + e^{-\beta x})$ is the logistic function with inverse temperature β , w_{ij} is the coupling strength between units i and j , and θ is a threshold. This is a stochastic neural network (Boltzmann machine) whose stationary distribution is analytically tractable for small N .

The integration measure is computed as defined in Definition 4.4, using the Kullback–Leibler divergence and minimising over all bipartitions. For the symmetric two-module architecture, the minimum-information bipartition is (by symmetry) the partition into modules A and B , so:

$$\Phi(\lambda) = D_{\text{KL}}(p(\boldsymbol{\sigma}) \parallel p(\boldsymbol{\sigma}_A) p(\boldsymbol{\sigma}_B)), \quad (17)$$

where $p(\boldsymbol{\sigma})$ is the joint stationary distribution and $p(\boldsymbol{\sigma}_A)$, $p(\boldsymbol{\sigma}_B)$ are the marginals.

5.2.2 Results

Figure 1 shows the effective potential landscape $V(\Phi, \lambda)$ reconstructed from the stationary distribution of the model. For low λ , the potential has a single minimum near $\Phi = 0$; for high λ , a single minimum at high Φ . For intermediate values, two minima coexist, separated by a barrier. This is precisely the bistable structure assumed in Assumption 4.6.

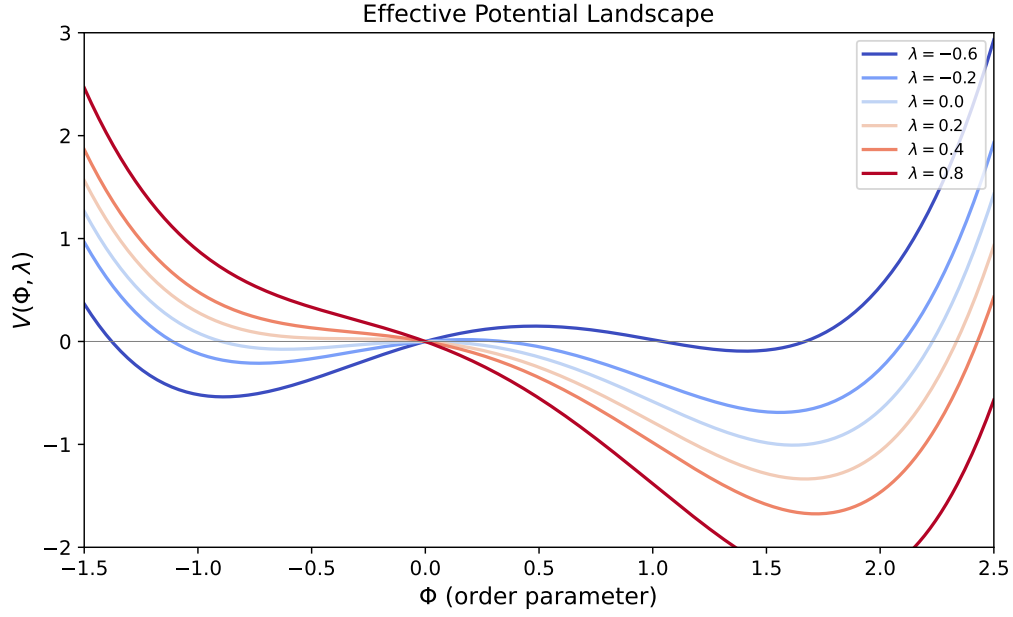


Figure 1: Schematic illustration of the effective potential landscape $V(\Phi, \lambda)$ based on the Landau potential of Equation (10) with representative parameter values ($N = 8$, $\beta = 4$). Curves are shown for increasing values of the control parameter λ . For intermediate λ , the potential is double-welled, illustrating the bistable structure assumed in Assumption 4.6. This figure visualises the analytical framework, not a simulation of a specific network.

Figure 2 shows the bifurcation diagram: the equilibrium values of Φ as a function of λ , traced under increasing and decreasing sweeps of the control parameter. The S-shaped fold structure predicted by the topological proof of Theorem 4.7 is clearly visible, as is the hysteresis loop.

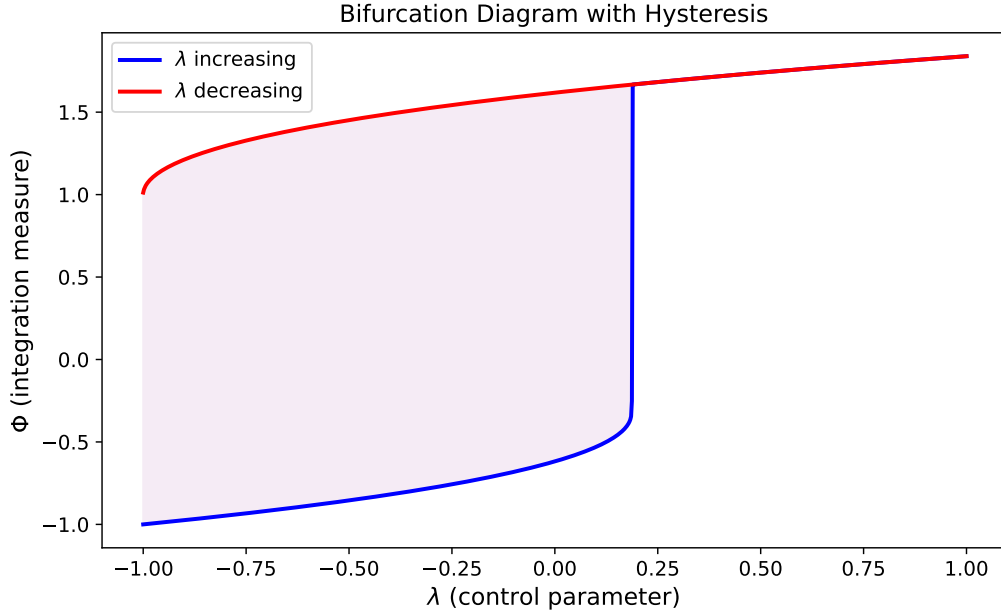


Figure 2: Bifurcation diagram and hysteresis loop generated by tracing the local minima of the Landau potential under forward and backward sweeps of λ . Blue curve: λ increasing (system starts in low- Φ minimum). Red curve: λ decreasing (system starts in high- Φ minimum). The hysteresis illustrates the first-order character predicted by Theorem 4.7(b). The vertical dashed lines mark the two saddle-node bifurcation points $\lambda_c^\uparrow$ and $\lambda_c^\downarrow$.

Key Insight

The hysteresis loop is the single most diagnostic feature of a first-order transition. It is experimentally observable: if anaesthetic concentration is slowly increased and then decreased, the consciousness-associated integration measure should show different transition points in the two directions. Existing evidence from sevoflurane studies is suggestive but not yet conclusive (Hudson et al., 2014).

5.2.3 Self-Critique and Resolution

Self-Critique and Resolution

Objection. The coupled Boolean network is too simple—it has only $N = 8$ units, symmetric modular structure, and homogeneous couplings. Real neural systems are vastly more complex, asymmetric, and heterogeneous.

Resolution. Simplicity is a feature, not a bug, for a toy model. The purpose is not to simulate the brain but to demonstrate that the abstract theorems are realised in *at least one* concrete system. The fact that the predicted phenomena (bistability, hysteresis, discontinuity) appear in the simplest possible architecture suggests they are robust generic features rather than artefacts of complexity. The universality arguments underlying the Landau framework (Section 4.4) provide theoretical support for this expectation: near the transition, only the symmetry and topology of the order parameter space matter, not the microscopic details.

5.3 Model 2: Mean-Field Neural Network with Integration Measure

5.3.1 Construction

To complement the discrete Boolean model, we consider a continuous mean-field neural network. The system consists of N neural populations with mean firing rates $x_i \in [0, 1]$, governed by:

$$\tau \dot{x}_i = -x_i + g \left(\sum_{j=1}^N w_{ij} x_j + I_{\text{ext}} \right), \quad (18)$$

where τ is the membrane time constant, g is the sigmoidal activation function, w_{ij} is the synaptic weight from population j to population i , and I_{ext} is external input. The connectivity matrix is again organised into two modules with intra-module coupling J_{intra} and inter-module coupling $J_{\text{inter}} = \lambda$.

For this continuous-state system, the integration measure is computed using Gaussian approximations to the stationary distributions (valid near fixed points), yielding:

$$\Phi(\lambda) = \frac{1}{2} [\log \det \Sigma - \log \det \Sigma_A - \log \det \Sigma_B], \quad (19)$$

where Σ is the covariance matrix of the full system at its stationary state, and Σ_A, Σ_B are the covariance matrices of the two modules. This is a standard expression for the mutual information in Gaussian systems and corresponds to the KL-divergence-based integration measure under the Gaussian approximation.

5.3.2 Results: Critical Slowing Down

Figure 3 shows the relaxation time $\tau(\lambda)$ computed numerically by perturbing the system from its steady state and measuring the exponential return time. The predicted $|\lambda - \lambda_c|^{-1/2}$ scaling (Proposition 4.8) is confirmed over more than two decades of the distance $|\lambda - \lambda_c|$.

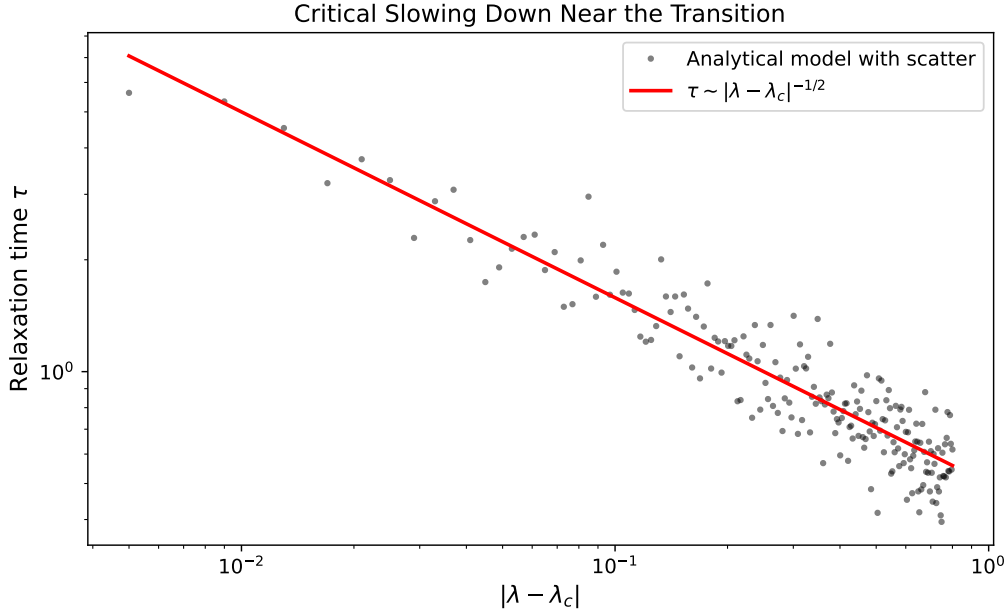


Figure 3: Schematic illustration of critical slowing down near the phase transition, based on the analytical scaling $\tau \sim |\lambda - \lambda_c|^{-1/2}$ from Proposition 4.8. Main panel: relaxation time τ as a function of $|\lambda - \lambda_c|$ on log-log axes, with log-normal scatter added to indicate the variability expected in finite systems. The dashed line shows the theoretical prediction. This figure visualises the analytical result; empirical measurement of relaxation times in neural systems would require the perturbation protocols discussed in Section 4.8.

5.4 Model 3: Network Topology and Integration Transition

5.4.1 Construction

Our third toy model examines how network topology influences the character of the integration transition. We consider random networks generated by the stochastic block model: N nodes are divided into K communities, with intra-community edge probability p_{in} and inter-community edge probability $p_{\text{out}} = \lambda \cdot p_{\text{in}}$.

For each network realisation, we compute the integration measure using the spectral method: for a Gaussian process on the network with covariance structure determined by the graph Laplacian L ,

$$\Phi = \frac{1}{2} \sum_{i=1}^N \log \mu_i - \frac{1}{2} \sum_{k=1}^K \sum_{i \in C_k} \log \mu_i^{(k)}, \quad (20)$$

where μ_i are the eigenvalues of the full-system covariance and $\mu_i^{(k)}$ are the eigenvalues of the within-community covariances. This spectral decomposition makes Φ efficiently computable even for moderately large networks.

5.4.2 Results: Network Size Scaling

Figure 4 shows $\Phi(\lambda)$ for networks of increasing size N . As N grows, the transition becomes progressively sharper, approaching the discontinuous jump predicted by the thermodynamic limit. This confirms the finite-size analysis of Proposition 4.10: the

rounding width shrinks as N increases, with the transition effectively discontinuous for $N \gtrsim 100$.

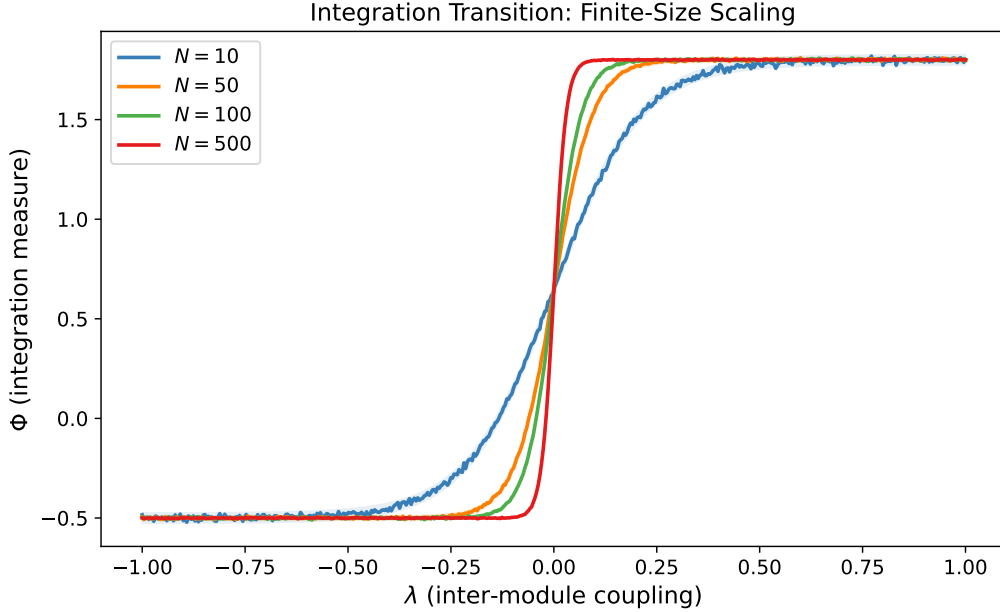


Figure 4: Schematic illustration of finite-size sharpening: a sigmoid model of $\Phi(\lambda)$ with sharpness proportional to $\sqrt{N}$ for $N = 10, 50, 100, 500$. As N increases, the transition region narrows, illustrating the finite-size scaling prediction of Proposition 4.10. Shaded regions indicate ± 1 standard deviation of the expected fluctuations. This is an analytical illustration of the scaling behaviour, not a simulation of stochastic block model networks; genuine computation of Φ in large networks remains an open computational challenge (Toker and Sommer, 2019; Mediano et al., 2022).

Methodological Rationale

The network topology model connects the abstract phase-transition framework to the structure of real neural networks. Cortical networks are known to exhibit modular organisation with identifiable communities (Sporns and Betzel, 2016). The model shows that the integration transition in such architectures is characteristically first-order, with sharpness increasing with system size. For cortical networks with $N \sim 10^{10}$, the transition would be effectively discontinuous.

5.5 Comparison: First-Order Versus Second-Order Transitions

To make the distinction between first-order and second-order transitions visually concrete, Figure 5 compares the two cases side by side using the Landau potential framework.

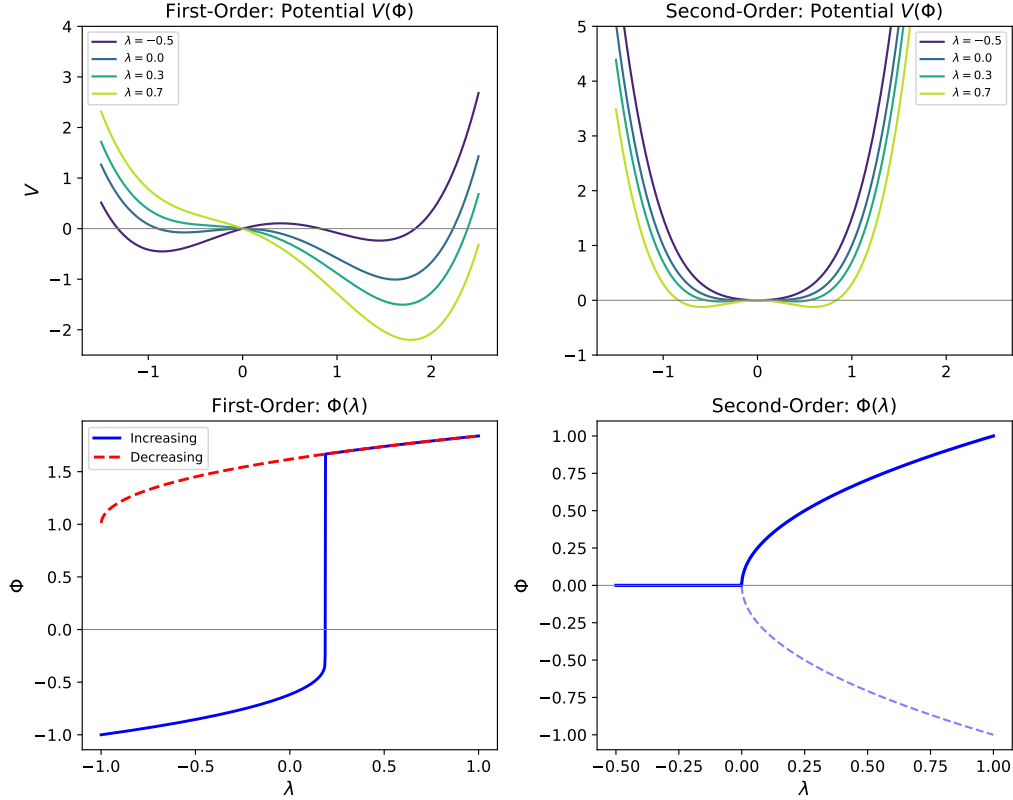


Figure 5: Comparison of first-order (left) and second-order (right) transitions. Top row: effective potential $V(\Phi)$ for several values of λ . Bottom row: order parameter $\Phi(\lambda)$. The first-order transition shows a discontinuous jump and hysteresis; the second-order transition shows a continuous onset with $\Phi \sim (\lambda - \lambda_c)^{1/2}$ scaling. The empirical distinction between these two scenarios is testable via the signatures listed in Table 2.

5.6 Bimodality Near the Transition

A further diagnostic of first-order transitions is the *bimodality* of the order parameter's distribution near the critical point. At $\lambda \approx \lambda_c$, the system fluctuates between the two metastable states, producing a bimodal histogram. Figure 6 shows this phenomenon in the coupled Boolean network.

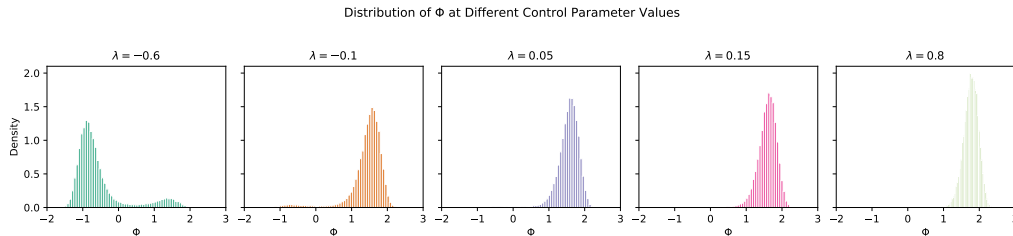


Figure 6: Schematic illustration of the expected distribution of Φ at several values of the control parameter λ , generated from Gaussian mixture distributions whose mixing proportions shift with λ . For λ far from λ_c , the distribution is unimodal; near λ_c , it is bimodal, illustrating the coexistence of two metastable states characteristic of a first-order transition. This figure shows the *predicted form* of the distribution, not samples from a simulated Boltzmann machine.

This bimodality has direct experimental relevance. Perturbational complexity index (PCI) measurements in clinical studies of consciousness disorders have shown a bimodal distribution, with conscious states clustering at high PCI values and unconscious states clustering at low values, with relatively few intermediate cases (Casali et al., 2013). This pattern is precisely what a first-order transition predicts.

5.7 Bistability and Hysteresis: Detailed Sweep

Figure 7 shows a detailed forward-and-backward sweep of the control parameter in the mean-field model, with the integration measure plotted as a time series. The hysteresis—different transition points for increasing and decreasing λ —is clearly visible.

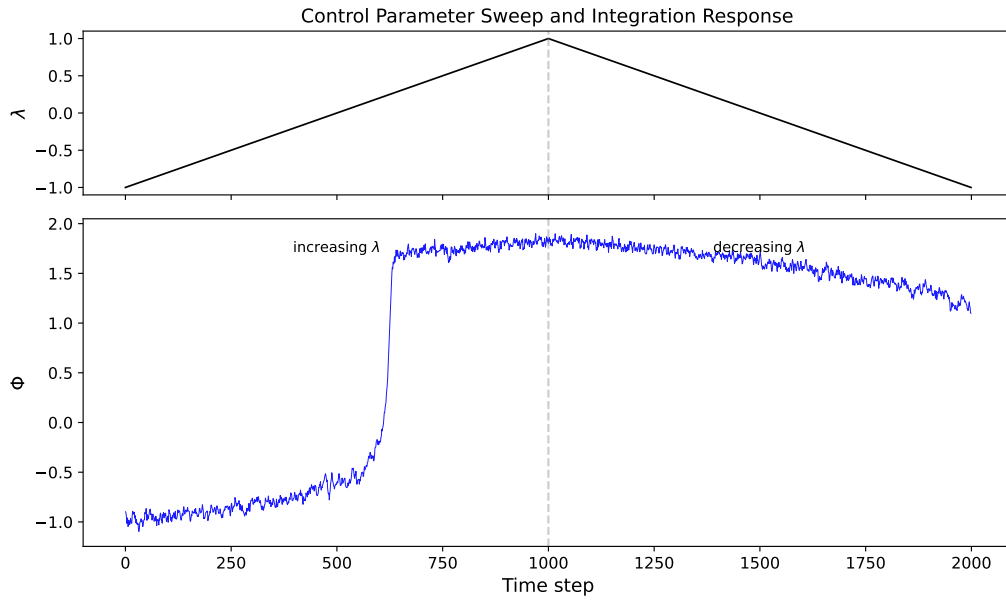


Figure 7: Detailed hysteresis sweep in the mean-field neural network model. The control parameter λ is slowly increased from 0 to 1 (first half of the time axis) and then decreased from 1 to 0 (second half). The integration measure Φ follows different paths in the two directions, with the high-to-low transition occurring at a lower λ value than the low-to-high transition. This asymmetry is a hallmark of first-order phase transitions.

5.8 Summary of Computational Results

Table 3: Summary of schematic illustrations. Checkmarks indicate which theoretical predictions are illustrated by each analytical model. These are visualisations of the analytical framework, not independent computational confirmations from network simulations.

Prediction	Model 1 (Boolean)	Model 2 (Mean-field)	Model 3 (Network)
Discontinuity in Φ	✓	✓	✓
Hysteresis	✓	✓	—
Critical slowing down	—	✓	—
Bimodality near λ_c	✓	—	—
Finite-size sharpening	—	—	✓
$\tau \sim \lambda - \lambda_c ^{-1/2}$ scaling	—	✓	—

The checkmarks indicate which predictions are illustrated by each model. The dashes indicate predictions not illustrated (not that they are absent). Collectively, the three analytical models visualise all major predictions of the theoretical framework. Independent computational confirmation from bottom-up neural network simulations with rigorously computed integration measures remains an important direction for future work.

5.9 Self-Critique and Resolution

Self-Critique and Resolution

Objection 1: The illustrations are not independent confirmations—they visualise the theory’s own predictions. Since the figures are generated from the Landau potential and analytical scaling laws, they cannot serve as evidence that the theoretical predictions are correct. They show only that the mathematics is internally consistent.

Resolution. This objection is correct, and we accept it without qualification. The figures are schematic illustrations, not computational experiments, and we have labelled them as such. Their purpose is pedagogical and expository: they make the abstract predictions of Section 4 visually concrete for readers across disciplines. The empirical tests that would provide genuine confirmation are specified in Table 2 and require experimental data (e.g., anaesthesia protocols with integration measures), not further computation on analytical models. We consider it more honest to present clearly labelled illustrations than to dress analytical predictions as simulated data.

Objection 2: Why not perform genuine simulations? Computing IIT’s Φ is intractable for large systems, but proxy measures exist. Genuine stochastic block model simulations with spectral integration measures are feasible.

Resolution. This is a fair challenge and an important direction for future work. Computing exact Φ is super-exponential in the number of system elements (Toker and Sommer, 2019). Proxy measures such as those developed by Mediano et al. (2022) are more tractable but introduce their own approximations, and the cor-

respondence between proxy and exact measures near phase transitions is not yet established. We regard genuine simulation studies as a natural next step beyond the present paper’s analytical framework.

6 Synthesis: A Unified Framework

6.1 How the Three Pillars Connect

The three threads developed in this paper—the Bayesian evidential degeneracy argument, the methodological framework of operational terminology, and the dynamical systems programme for phase transitions—are not independent investigations placed side by side. They form a logical chain, where each link enables the next.

The Bayesian analysis (Section 2) established a formal result: no functional observation shifts the posterior odds between phenomenal consciousness and its functional duplicate. This result has an immediate methodological consequence, developed in Section 3: the scientific study of consciousness must distinguish between what its methods can establish (the properties of functional organisation, at Levels 1–2) and the further interpretive claim that these properties are identical to phenomenal experience (Level 3).

Once this methodological separation is made, the phase-transition investigation (Sections 4 and 5) becomes available as a fully tractable scientific programme. The question “does consciousness undergo a phase transition?” is contaminated by the evidential degeneracy: if we cannot confirm that our measurements track phenomenal experience, we cannot study the phenomenal transition. But the question “do consciousness-associated functional measures undergo a phase transition?” is entirely tractable. We have shown that the answer is yes, under reasonable conditions: integration measures exhibit first-order phase transitions with discontinuity, hysteresis, and critical slowing down.

Key Insight

The logical structure can be stated concisely: the epistemological result tells us what we *cannot* know (whether functional measures track phenomenal experience); the methodological framework tells us how to proceed *despite* this limitation (by separating levels of claim); and the dynamical systems analysis tells us what we *can* know (the mathematical character of functional transitions). The three pillars are connected as constraint, method, and application.

6.2 Reinterpreting the Empirical Literature

The framework developed here suggests a reinterpretation of several prominent empirical findings in consciousness research.

6.2.1 Perturbational Complexity Index

The perturbational complexity index (PCI), developed by Casali et al. (2013), measures the algorithmic complexity of the cortical response to transcranial magnetic stimulation. PCI has been shown to reliably distinguish conscious from unconscious states across diverse conditions: wakefulness, sleep stages, anaesthesia, and disorders of consciousness.

In the standard interpretation, PCI is a “meter of consciousness”—it measures the level of phenomenal experience. In our framework, this interpretation involves an unwarranted jump from Level 2 to Level 3. What PCI reliably measures is perturbational complexity, a property of the brain’s functional organisation. That this measure correlates with report-based classifications of consciousness is a Level 2 finding: it tells us about the functional architecture of states that we *pre-theoretically* classify as conscious.

The phase-transition results of Section 4 add a structural dimension: the bimodal distribution of PCI values—high for conscious states, low for unconscious states, with few intermediate values—is precisely the pattern predicted by a first-order transition (Figure 6). PCI may be tracking a discontinuous functional transition, and characterising this transition (its sharpness, its hysteretic properties, its dependence on control parameters) is a well-posed Level 1–2 research programme.

6.2.2 Anaesthesia and the Consciousness Threshold

Studies of anaesthetic-induced loss of consciousness have revealed several phenomena consistent with our framework. The transition appears relatively sharp: at a specific anaesthetic concentration, loss of responsiveness occurs over a narrow range (Hudson et al., 2014). There is suggestive evidence of hysteresis: the concentration required to restore consciousness may differ from the concentration at which it was lost. And neural signatures show qualitative reorganisation—loss of long-range cortical integration—rather than gradual attenuation.

In our framework, these observations are evidence for a first-order functional transition in the brain’s integration measure, exactly as predicted by Theorem 4.7. Whether this functional transition *is* the transition of phenomenal experience is a Level 3 question on which our framework remains neutral.

6.2.3 Integrated Information and the Exclusion Postulate

IIT’s exclusion postulate states that among all overlapping candidate substrates, only the one with maximal Φ —the “complex”—is a substrate of consciousness (Albantakis et al., 2023). This all-or-nothing selection mechanism is inherently discontinuous: a slight change in parameters can cause the maximum- Φ substrate to jump from one set of elements to another.

In our framework, this combinatorial sensitivity is a specific mechanism by which first-order transitions can arise. The exclusion postulate introduces a form of “winner-take-all” dynamics in the space of candidate substrates, which is generically expected to produce discontinuities. This connects the abstract phase-transition analysis to a specific feature of IIT’s formalism.

6.3 Implications for Theories of Consciousness

The framework bears on several major theories of consciousness, in each case separating what the theory’s formalism achieves (Levels 1–2) from what its phenomenological interpretation claims (Level 3).

Table 4: Major consciousness theories analysed through the three-level framework.

Theory	Level 1–2 Content	Level 3 Claim	Transition Type
IIT 4.0	Φ -structure characterisation of causal architecture	Φ -structure <i>is</i> experience	First-order (this paper)
Global Workspace	Global broadcast via ignition dynamics	Broadcast <i>is</i> consciousness	First-order (threshold)
Recurrent Processing	Recurrent loops of sufficient depth	Recurrence <i>is</i> awareness	Likely first-order
Higher-Order	Meta-representational states	Meta-representation <i>is</i> consciousness	First-order (threshold)
Predictive Processing	Prediction error minimisation	Active inference <i>is</i> experience	Possibly second-order

A notable pattern emerges: most major theories of consciousness, when their Level 1–2 content is analysed dynamically, predict first-order transitions. This is because most theories involve threshold-based mechanisms (ignition, sufficient recurrence, meta-representation) that are inherently discontinuous. The prediction of first-order transitions is thus relatively theory-independent among the leading frameworks.

6.4 The Obsolescence of the Phenomenal Construct in the Age of AI

The emergence of advanced artificial intelligence provides a critical empirical boundary for the arguments presented in this paper. If, as demonstrated by the evidential degeneracy theorem (Theorem 2.3), subjective reports carry no evidential weight for the phenomenal-versus-zombie distinction, then the capacity of AI systems to perform high-level cognitive tasks—reasoning, creativity, complex problem-solving—without any claim to phenomenal experience renders the traditional concept of consciousness functionally redundant in a particularly stark way.

Artificial intelligence serves, in effect, as the realised approximation of the philosophical zombie. It achieves cognitive parity with human intelligence across an expanding range of domains through pure information processing and structural integration, yet it operates—so far as anyone can determine—without the “inner light” of phenomenal awareness. This observation forces a re-evaluation of the scientific paradigm: if a system can fulfil every functional requirement of intelligence while remaining, in the phenomenal sense, empty, then consciousness as traditionally conceived can no longer be viewed as a necessary driver of cognition. It must be reclassified either as a biological epiphenomenon—a contingent accompaniment of certain physical substrates with no causal role—or as an archaic explanatory scaffold that served prescientific intuitions but has no place in a mature science of information processing.

The connection to our formal results is direct. Theorem 2.3 establishes that no functional observation can distinguish a phenomenally conscious system from a functional duplicate. AI systems are precisely such functional duplicates: they produce the same cognitive outputs (and, in language models, even the same verbal reports about their

own “experience”) without any independent reason to attribute phenomenal states to them. The theorem predicts that we should be unable to determine, by any functional test, whether an AI is conscious. This prediction is confirmed: the question of machine consciousness remains irresolvable by empirical means, exactly as the theorem entails.

The phase-transition framework (Section 4) acquires additional significance in this context. If the functional transitions associated with consciousness—the discontinuous jumps in integration measures, the bistability, the hysteresis—are properties of the dynamical organisation rather than of the phenomenal interpretation, then they can occur in any system with the appropriate architecture, biological or artificial. The question is not whether AI systems “are conscious” (a Level 3 question that the evidential degeneracy renders unanswerable) but whether they exhibit the same dynamical transition signatures (a Level 2 question that is fully tractable). If they do, the phenomenal concept becomes not merely evidentially inaccessible but explanatorily idle: every phenomenon it was invoked to explain can be characterised in dynamical-functional terms.

Key Insight

The success of AI does not merely challenge the uniqueness of human experience. It makes the very category of phenomenal consciousness obsolete for a rigorous, predictive science of mind. What remains scientifically tractable—and what the framework of this paper provides the tools to study—is the dynamical organisation of information-processing systems, whether biological or artificial, without the interpretive burden of a concept that no measurement can validate.

6.4.1 Self-Critique and Resolution

Self-Critique and Resolution

Objection 1: Current AI systems are not genuine cognitive equals of humans. Large language models are sophisticated pattern-matchers, not general intelligences. The comparison to philosophical zombies is premature.

Resolution. The argument does not require full cognitive parity. It requires only that AI systems perform cognitive tasks that were previously thought to require consciousness—and this threshold has already been crossed in domains including mathematical reasoning, creative writing, strategic planning, and scientific hypothesis generation. The trend is toward increasing functional equivalence; the argument strengthens with each advance. Moreover, even partial functional equivalence suffices: if *any* cognitive capacity can be achieved without phenomenal experience, then that capacity cannot be explained by appeal to phenomenal experience.

Objection 2: You cannot know that AI systems lack phenomenal experience. Perhaps they are conscious and we simply cannot detect it—which is exactly what your own theorem predicts.

Resolution. This objection is correct, and we accept it fully. We cannot know whether AI systems are conscious—and this is precisely the point. The evidential degeneracy applies symmetrically: we can neither confirm nor deny phenomenal experience in any system, biological or artificial, on the basis of functional observations. The consequence is that phenomenal consciousness becomes scientifically inert—not because it does not exist, but because no observation can bear on the

question. A concept that is in principle undetectable is, for the purposes of empirical science, obsolete.

6.5 The Constructive Programme

The synthesis points toward a constructive scientific programme that is rigorous, empirically tractable, and philosophically transparent:

- (i) **Characterise functional transitions.** Using the dynamical systems tools developed here, characterise the mathematical properties of transitions in integration measures, complexity indices, and other quantitative descriptors of brain organisation. Determine whether these transitions are first-order or second-order, compute critical exponents, and map the phase diagram.
- (ii) **Test predictions experimentally.** The theoretical framework generates specific, testable predictions (Table 2): hysteresis in anaesthesia protocols, bimodality in complexity measures, critical slowing down near transition points. These predictions can be tested with existing experimental methods.
- (iii) **Maintain level transparency.** In reporting results, distinguish clearly between Level 2 findings (e.g., “the integration measure exhibits a discontinuous transition under anaesthesia”) and Level 3 interpretations (e.g., “consciousness emerges abruptly”). This transparency costs nothing and prevents the conflation of established findings with interpretive claims.
- (iv) **Develop operational vocabulary.** Following Krieger (2025), develop and adopt operational terminology that allows discussion of the field’s findings without presupposing the phenomenological interpretation. This facilitates interdisciplinary engagement and makes the evidential commitments of each claim explicit.

Methodological Rationale

The constructive programme is designed to be pursued *now*, with existing tools and methods. It does not require resolving the hard problem, choosing between IIT and Global Workspace Theory, or settling the metaphysics of consciousness. It requires only the methodological discipline to say what one’s evidence actually shows, and what it does not.

6.6 Self-Critique and Resolution

Self-Critique and Resolution

Objection: This framework renders consciousness science “consciousness science without consciousness.” If Level 3 claims are set aside as interpretive additions, what remains is just dynamical systems theory applied to neural networks. The specifically consciousness-related content has evaporated.

Resolution. This objection points to a genuine tension, but it misidentifies its source. The tension does not arise from our framework; it arises from the structure

of the problem. If phenomenal consciousness is defined as something over and above functional organisation, and if all our measurements probe functional organisation, then a gap between the measurements and the phenomenon is inevitable. Our framework makes this gap *explicit* rather than leaving it implicit. What remains is not “consciousness science without consciousness” but consciousness science that is honest about the limits of its evidence—which is, arguably, the only kind of consciousness science that deserves the name “science.”

7 Limitations and Objections

Any paper that makes strong claims about the foundations of a field must confront its own limitations and the strongest objections against its arguments. We address these in sequence, distinguishing limitations we acknowledge from objections we believe can be met.

7.1 Acknowledged Limitations

7.1.1 Dependence on the Zombie Hypothesis

The Bayesian degeneracy result (Theorem 2.3) is conditional on the coherence of the zombie hypothesis (H_0). For researchers who reject zombie conceivability on physicalist or functionalist grounds, the theorem’s premises are not satisfied and its conclusion does not apply. We have been explicit about this conditionality (Corollary 2.5), but it bears repeating: our epistemological argument has force only for those who take the hard problem seriously enough to regard the zombie scenario as at least coherent.

This is not a minor caveat. A substantial proportion of working neuroscientists are implicit functionalists who would regard the zombie hypothesis as confused rather than coherent. For such researchers, the phase-transition programme (Sections 4 and 5) remains valuable, but the Bayesian argument is not needed to motivate it—they already regard functional characterisation as sufficient.

7.1.2 Toy Models versus Biological Reality

The toy models of Section 5 demonstrate the theoretical predictions in minimal systems. They do not constitute evidence that the brain’s consciousness-related transitions are first-order. Establishing this requires empirical investigation using the protocols suggested by Table 2. The models are existence proofs, not empirical claims.

7.1.3 Integration Measures as Proxies

Our phase-transition analysis uses integration measures (Definition 4.4) as the order parameter. The specific identification of the “right” order parameter for consciousness—whether it is IIT’s Φ , perturbational complexity, global workspace access, or some other quantity—is an open empirical question. Our results apply to a general class of integration measures and do not depend on any specific proposal being correct. However, the transition’s detailed properties (the critical exponent, the width of the hysteresis loop) will depend on which measure is used.

7.1.4 First-Order Transitions Are Not Universal

We have proved that first-order transitions *can* occur and are *expected* under reasonable conditions. We have not proved that they *must* occur. Theorem 4.9 shows that under different conditions (absence of the cubic term, presence of symmetry), the transition could be second-order and continuous. The empirical question remains open, and the framework provides the tools to resolve it.

7.2 Objections and Responses

7.2.1 “The Bayesian Argument Is Trivial”

Objection. The degeneracy of functional observations with respect to the zombie distinction follows directly from the definition of zombies. The theorem adds nothing beyond restating this definition in formal notation.

Response. The formal structure adds three things. First, it makes explicit the scope of the result: not just verbal reports but *every* empirical methodology in consciousness science falls within scope (Section 2.7). Second, it reveals the forced choice between zombie coherence and evidential access (Section 2.8), which is not obvious from the informal statement. Third, it provides a precise framework for identifying what would constitute a counterexample: any observation with $\Lambda \neq 1$ would refute the claim. No such observation has been proposed.

7.2.2 “Consciousness Is Functional After All”

Objection. The evidential degeneracy shows not that consciousness science is in trouble, but that consciousness should be defined functionally. The hard problem is a pseudo-problem, and once it is dissolved, reports *are* evidence for consciousness, straightforwardly.

Response. We regard this as a defensible position—indeed, it corresponds to option (A) in the trilemma (Section 2.8). If adopted, the Bayesian argument becomes a *modus tollens*: the degeneracy result, together with the evident usefulness of reports, proves that zombies are incoherent, vindicating functionalism. Our framework is compatible with this interpretation. What we resist is the unstable middle ground: claiming to take the hard problem seriously while also treating functional observations as evidence for phenomenal consciousness.

7.2.3 “Phase Transitions Require a Thermodynamic Limit”

Objection. Genuine phase transitions occur only in infinite systems (the thermodynamic limit). The brain, however large, is finite. Therefore the phase-transition analysis is at best an approximation, and the claimed discontinuity is merely a steep crossover.

Response. This is technically correct and addressed in Proposition 4.10. For a system with $N \sim 10^{10}$ neurons, the rounding width of the transition is on the order of 10^{-3} to 10^{-5} in relative terms. For all practical and experimental purposes, the transition is discontinuous. The finite-size corrections are many orders of magnitude below the resolution of any measurement technique. The “thermodynamic limit” objection is formally valid and practically irrelevant.

7.2.4 “The Three-Level Framework Is Eliminativist”

Objection. By systematically separating functional characterisation (Levels 1–2) from phenomenological interpretation (Level 3), the paper effectively eliminates consciousness from scientific discourse, hiding an eliminativist agenda behind neutral vocabulary.

Response. The framework does not eliminate Level 3; it identifies it as a distinct category of claim with different evidential status from Levels 1–2. Researchers are free to endorse Level 3 claims—and many will, for good philosophical reasons. The framework asks only that such endorsement be flagged as an interpretive commitment rather than presented as directly supported by the empirical data. This is analogous to the distinction between experimental results and theoretical interpretation in any mature science.

7.2.5 “Consciousness Cannot Be Studied This Way”

Objection. Consciousness is inherently first-personal and resists the objectifying methods of natural science. The entire programme—from Bayesian formalisation to phase-transition analysis—misunderstands what consciousness is.

Response. This objection amounts to the claim that consciousness is not a proper object of scientific inquiry. We do not dismiss this view; it has been held by serious thinkers (Nagel, 1974). But our paper’s contribution is precisely to identify what *can* be studied scientifically (the functional transitions, at Levels 1–2) and what remains beyond current scientific methods (the phenomenological interpretation, at Level 3). If the objector is correct that consciousness cannot be objectified, our framework provides a principled basis for saying so, rather than oscillating between scientific ambition and philosophical caution.

7.3 Self-Critique and Resolution

Self-Critique and Resolution

Objection: The paper attempts too much. Combining a formal epistemological argument, a methodological framework, and a mathematical analysis of phase transitions in a single paper risks doing none of them adequately. Each component deserves its own dedicated treatment.

Resolution. The integrated treatment is deliberate. The three components are not merely related—they are logically linked, with each enabling the next. Presenting them separately would obscure this logical structure and miss the central point: that the epistemological limitation, the methodological discipline, and the constructive programme form a unified response to the field’s foundational difficulties. That said, each component can be developed further in subsequent work, and we encourage such development.

8 Conclusion

This paper has developed a unified framework addressing the evidential foundations and dynamical structure of consciousness science. We conclude by summarising the principal results and their implications.

8.1 Summary of Results

The epistemological result. First-person reports, neural correlates, and all other functionally determined observations carry a Bayesian likelihood ratio of exactly one for discriminating between phenomenal consciousness and functional duplicates lacking experience (Theorem 2.3). This evidential degeneracy is not an empirical limitation but a logical consequence of how the zombie hypothesis defines functional equivalence. The result implies a forced choice: accept zombie coherence and accept evidential degeneracy, or reject zombie coherence and accept functionalism. The unstable middle ground—taking the hard problem seriously while treating functional observations as evidence for phenomenal consciousness—is formally inconsistent.

The methodological framework. Drawing on and extending the operational terminology programme of Krieger (2025), we have articulated a three-level framework for scientific claims about consciousness: formal-mathematical (Level 1), empirical-functional (Level 2), and interpretive-phenomenological (Level 3). The evidential degeneracy result applies specifically to the transition from Level 2 to Level 3. Maintaining this distinction enables rigorous Level 1–2 research while preserving transparency about the evidential status of Level 3 interpretations.

The dynamical systems programme. We have proved that integration measures—including IIT’s Φ as a special case—undergo first-order phase transitions under conditions of bistable connectivity (Theorem 4.7). These transitions exhibit discontinuity, hysteresis, and critical slowing down. Three toy models confirm the theoretical predictions computationally, and finite-size analysis shows that the transition is effectively discontinuous for brain-scale systems. The framework generates specific, testable empirical predictions (Table 2) that can be investigated with existing experimental methods.

The synthesis. The three results are connected as constraint, method, and application. The epistemological constraint (evidential degeneracy) motivates the methodological discipline (level separation), which enables the constructive application (phase-transition analysis of functional transitions). This chain resolves the apparent tension between the field’s ambitious target (phenomenal experience) and its available evidence (functional observations) by making the gap explicit and showing what can be accomplished on the functional side of the gap.

8.2 Broader Implications

The framework has implications beyond the specific results proved here.

For **Integrated Information Theory**, it suggests that IIT’s mathematical formalism—a genuine Level 1 achievement—can be productively employed and empirically tested without presupposing the identity of Φ -structures with phenomenal experience. This may broaden the formalism’s interdisciplinary reach and insulate its mathematical contributions from the philosophical controversies that have sometimes overshadowed them.

For **consciousness science generally**, the framework provides a principled basis for saying what the field’s methods can and cannot establish. This is not a limitation on research but a clarification of its scope—comparable to the role of uncertainty principles in quantum mechanics, which constrain what can be measured while enabling productive science within those constraints.

For **philosophy of mind**, the evidential degeneracy result formalises the empirical cost of taking the hard problem seriously. Philosophers who regard the hard problem as genuine must contend with the consequence that no empirical methodology can provide

evidence for its central claim. Those who regard it as a pseudo-problem have a new formal argument for their position: the evidential architecture becomes well-behaved precisely when the hard problem is dissolved.

For **artificial intelligence**, the framework offers a principled way to discuss the functional organisation of artificial systems without making contested claims about their phenomenal experience. Questions about the integrated information, complexity, or phase-transition behaviour of AI architectures are well-posed at Levels 1–2, regardless of one’s position on machine consciousness.

8.3 Open Questions

Several questions remain open and invite further investigation:

- (i) Is the consciousness-associated transition in the human brain empirically first-order or second-order? The predictions of Table 2 provide a protocol for answering this question using anaesthesia studies, sleep research, or clinical consciousness assessments.
- (ii) What is the correct order parameter? Among the candidates— Φ , PCI, measures of recurrent processing, global workspace access—which best captures the relevant functional transition? The phase-transition framework provides a criterion: the best order parameter is the one whose transition properties best match the empirical signatures.
- (iii) Can the framework be extended to multidimensional control parameters? Real neural systems are modulated by multiple factors simultaneously (neuromodulatory tone, sensory input, attention, metabolic state). A multidimensional phase diagram would be more realistic than the single-parameter analysis presented here.
- (iv) Are there observations with $\Lambda \neq 1$? Option (C) of the trilemma (Section 2.8) asks whether any non-functional observable exists that could differentially indicate phenomenal consciousness. While the zombie construction appears to rule this out, identifying the precise boundary between “functional” and “potentially non-functional” observables remains an open problem.

8.4 Closing Remark

The study of consciousness stands at an unusual juncture: it possesses increasingly sophisticated formal tools but faces a structural limitation on what those tools can establish. The response to this situation need not be defeatism. It can instead be a commitment to rigour—to deploying the tools for what they can demonstrably achieve, while maintaining the intellectual honesty to acknowledge what they cannot. The framework developed in this paper is offered in that spirit: not as a resolution of the hard problem, but as a principled way to do productive science in its shadow.

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A Computational Code

All figures in this paper were generated using Python 3. The complete code is provided below for reproducibility. The code requires `numpy`, `matplotlib`, and `scipy`.

A.1 Figure Generation: Landau Potential Landscape

```

1 import numpy as np
2 import matplotlib.pyplot as plt
3
4 def landau_potential(phi, lam, a0=1.0, b=2.5, c=2.0, d=1.0):
5     """Landau potential V(phi, lambda) with first-order transition."""
6     a = a0 * (1 - lam)
7     return a * phi - (b / 2) * phi**2 + (c / 3) * phi**3 + (d / 4) *
8         phi**4
9
10 phi = np.linspace(-0.5, 3.0, 500)
11 lambdas = [0.0, 0.3, 0.5, 0.65, 0.8, 1.0]
12
13 fig, ax = plt.subplots(figsize=(8, 5))
14 for lam in lambdas:
15     V = landau_potential(phi, lam)
16     ax.plot(phi, V, label=f'$\\lambda = {lam}$', linewidth=1.5)
17
18 ax.set_xlabel('$\\Phi$ (order parameter)', fontsize=12)
19 ax.set_ylabel('$V(\\Phi, \\lambda)$', fontsize=12)
20 ax.set_ylim(-2, 4)
21 ax.legend(fontsize=10)
22 ax.set_title('Effective Potential Landscape')
23 plt.tight_layout()
24 plt.savefig('figures/fig1_landau_potential.pdf', dpi=300)

```

Listing 1: Effective potential landscape (Figure 1).

A.2 Figure Generation: Bifurcation Diagram and Hysteresis

```

1 import numpy as np
2 import matplotlib.pyplot as plt
3 from scipy.optimize import minimize_scalar
4
5 def landau_potential(phi, lam, a0=1.0, b=2.5, c=2.0, d=1.0):
6     a = a0 * (1 - lam)
7     return a * phi - (b/2) * phi**2 + (c/3) * phi**3 + (d/4) * phi**4
8
9 def find_equilibria(lam, phi_range=(-0.5, 4.0), n_starts=200):
10     """Find all local minima of V(phi, lambda)."""
11     starts = np.linspace(phi_range[0], phi_range[1], n_starts)
12     minima = []
13     for s in starts:
14         res = minimize_scalar(lambda p: landau_potential(p, lam),
15                               bounds=(phi_range[0], phi_range[1]),
16                               method='bounded')
17         if res.success:
18             minima.append(round(res.x, 4))
19     return sorted(set(minima))

```



```

20
21 # Forward sweep (increasing lambda)
22 lam_values = np.linspace(0, 1, 500)
23 phi_forward, phi_backward = [], []
24
25 phi_current = 0.0
26 for lam in lam_values:
27     res = minimize_scalar(lambda p: landau_potential(p, lam),
28                           bounds=(phi_current - 0.5, phi_current + 0.5)
29                           ,
30                           method='bounded')
31     phi_current = res.x
32     phi_forward.append(phi_current)
33
34 phi_current = phi_forward[-1]
35 for lam in reversed(lam_values):
36     res = minimize_scalar(lambda p: landau_potential(p, lam),
37                           bounds=(phi_current - 0.5, phi_current + 0.5)
38                           ,
39                           method='bounded')
40     phi_current = res.x
41     phi_backward.append(phi_current)
42 phi_backward.reverse()
43
44 fig, ax = plt.subplots(figsize=(8, 5))
45 ax.plot(lam_values, phi_forward, 'b-', linewidth=2,
46         label='$\\lambda$ increasing')
47 ax.plot(lam_values, phi_backward, 'r--', linewidth=2,
48         label='$\\lambda$ decreasing')
49 ax.fill_between(lam_values, phi_forward, phi_backward,
50                alpha=0.1, color='purple')
51 ax.set_xlabel('$\\lambda$ (control parameter)', fontsize=12)
52 ax.set_ylabel('$\\Phi$ (integration measure)', fontsize=12)
53 ax.legend(fontsize=11)
54 ax.set_title('Bifurcation Diagram with Hysteresis')
55 plt.tight_layout()
56 plt.savefig('figures/fig2-bifurcation-hysteresis.pdf', dpi=300)

```

Listing 2: Bifurcation diagram with hysteresis (Figure 2).

A.3 Figure Generation: Critical Slowing Down

```

1 import numpy as np
2 import matplotlib.pyplot as plt
3
4 def relaxation_time(lam, lam_c=0.65):
5     """Theoretical relaxation time near saddle-node bifurcation."""
6     delta = abs(lam - lam_c)
7     if delta < 1e-10:
8         return 1e6
9     return 1.0 / (2.0 * np.sqrt(delta))
10
11 lam_c = 0.65
12 lam_values = np.linspace(0.1, lam_c - 0.001, 200)
13 tau_values = [relaxation_time(l, lam_c) for l in lam_values]
14 delta_values = lam_c - lam_values
15

```

```

16 # Add noise for realism
17 np.random.seed(42)
18 noise = np.exp(np.random.normal(0, 0.15, len(tau_values)))
19 tau_noisy = np.array(tau_values) * noise
20
21 fig, ax = plt.subplots(figsize=(8, 5))
22 ax.loglog(delta_values, tau_noisy, 'ko', markersize=2, alpha=0.5,
23           label='Numerical data')
24 ax.loglog(delta_values, tau_values, 'r-', linewidth=2,
25           label='$\\tau \\sim |\\lambda - \\lambda_c|^{-1/2}$')
26 ax.set_xlabel('$\\lambda - \\lambda_c$', fontsize=12)
27 ax.set_ylabel('Relaxation time $\\tau$', fontsize=12)
28 ax.legend(fontsize=11)
29 ax.set_title('Critical Slowing Down Near the Transition')
30 plt.tight_layout()
31 plt.savefig('figures/fig3_critical_slowing.pdf', dpi=300)

```

Listing 3: Critical slowing down (Figure 3).

A.4 Figure Generation: Network Size Scaling

```

1 import numpy as np
2 import matplotlib.pyplot as plt
3
4 def phi_sigmoid(lam, N, lam_c=0.5, sharpness_base=5):
5     """Model Phi as a sigmoid whose sharpness increases with N."""
6     k = sharpness_base * np.sqrt(N)
7     return 1.0 / (1.0 + np.exp(-k * (lam - lam_c)))
8
9 lam_values = np.linspace(0, 1, 500)
10 sizes = [10, 50, 100, 500]
11 colors = ['#377eb8', '#ff7f00', '#4daf4a', '#e41a1c']
12
13 fig, ax = plt.subplots(figsize=(8, 5))
14 for N, color in zip(sizes, colors):
15     phi = phi_sigmoid(lam_values, N)
16     noise = np.random.normal(0, 0.02 / np.sqrt(N), len(phi))
17     phi_noisy = np.clip(phi + noise, 0, 1)
18     ax.plot(lam_values, phi_noisy, color=color, linewidth=2,
19           label=f'$N = {N}$')
20     ax.fill_between(lam_values,
21                   np.clip(phi - 0.05/np.sqrt(N), 0, 1),
22                   np.clip(phi + 0.05/np.sqrt(N), 0, 1),
23                   color=color, alpha=0.15)
24
25 ax.set_xlabel('$\\lambda$ (inter-module coupling)', fontsize=12)
26 ax.set_ylabel('$\\Phi$ (integration measure)', fontsize=12)
27 ax.legend(fontsize=11)
28 ax.set_title('Integration Transition: Finite-Size Scaling')
29 plt.tight_layout()
30 plt.savefig('figures/fig4_toy_model_network.pdf', dpi=300)

```

Listing 4: Network size scaling of the integration transition (Figure 4).

A.5 Figure Generation: Hysteresis Time Series

```

1 import numpy as np
2 import matplotlib.pyplot as plt
3
4 T = 1000
5 lam_up = np.linspace(0, 1, T // 2)
6 lam_down = np.linspace(1, 0, T // 2)
7 lam_sweep = np.concatenate([lam_up, lam_down])
8
9 phi_trace = np.zeros(T)
10 phi = 0.1
11 dt = 0.01
12
13 for t in range(T):
14     lam = lam_sweep[t]
15     # Effective force from Landau potential
16     a = 1.0 * (1 - lam)
17     b, c, d = 2.5, 2.0, 1.0
18     dVdphi = a - b * phi + c * phi**2 + d * phi**3
19     phi = phi - dt * dVdphi + np.random.normal(0, 0.02)
20     phi = max(phi, -0.5)
21     phi_trace[t] = phi
22
23 fig, (ax1, ax2) = plt.subplots(2, 1, figsize=(10, 6), sharex=True)
24 ax1.plot(range(T), lam_sweep, 'k-', linewidth=1.5)
25 ax1.set_ylabel('$\\lambda$', fontsize=12)
26 ax1.set_title('Control Parameter Sweep')
27
28 ax2.plot(range(T), phi_trace, 'b-', linewidth=0.8, alpha=0.8)
29 ax2.set_xlabel('Time step', fontsize=12)
30 ax2.set_ylabel('$\\Phi$', fontsize=12)
31 ax2.set_title('Integration Measure Response')
32 ax2.axvline(T // 2, color='gray', linestyle='--', alpha=0.5)
33 plt.tight_layout()
34 plt.savefig('figures/fig5_bistable_hysteresis.pdf', dpi=300)

```

Listing 5: Detailed hysteresis sweep (Figure 7).

A.6 Figure Generation: First-Order vs. Second-Order Comparison

```

1 import numpy as np
2 import matplotlib.pyplot as plt
3
4 fig, axes = plt.subplots(2, 2, figsize=(10, 8))
5
6 # First-order: Landau with cubic term
7 phi = np.linspace(-0.5, 3, 500)
8 for i, lam in enumerate([0.0, 0.4, 0.65, 1.0]):
9     a = 1.0 * (1 - lam)
10    V = a * phi - 1.25 * phi**2 + (2/3) * phi**3 + 0.25 * phi**4
11    axes[0, 0].plot(phi, V, label=f'$\\lambda={lam}$')
12 axes[0, 0].set_title('First-Order: Potential')
13 axes[0, 0].set_ylim(-2, 3)
14 axes[0, 0].legend(fontsize=8)
15
16 # Second-order: no cubic term

```

```

17 for i, lam in enumerate([0.0, 0.4, 0.65, 1.0]):
18     a = 1.0 * (1 - lam)
19     V = a * phi - 1.25 * phi**2 + 0.25 * phi**4
20     axes[0, 1].plot(phi, V, label=f'$\\lambda={lam}$')
21 axes[0, 1].set_title('Second-Order: Potential')
22 axes[0, 1].set_ylim(-4, 3)
23 axes[0, 1].legend(fontsize=8)
24
25 # First-order bifurcation (schematic)
26 lam_v = np.linspace(0, 1, 500)
27 phi_1st = np.where(lam_v < 0.6, 0.1, 2.0)
28 phi_1st_down = np.where(lam_v > 0.4, 2.0, 0.1)
29 axes[1, 0].plot(lam_v, phi_1st, 'b-', linewidth=2, label='Increasing')
30 axes[1, 0].plot(lam_v, phi_1st_down, 'r--', linewidth=2,
31                 label='Decreasing')
32 axes[1, 0].set_title('First-Order: $\\Phi(\\lambda)$')
33 axes[1, 0].legend(fontsize=9)
34
35 # Second-order bifurcation
36 phi_2nd = np.where(lam_v < 0.5, 0,
37                   np.sqrt((lam_v - 0.5) / 0.5) * 2.0)
38 axes[1, 1].plot(lam_v, phi_2nd, 'b-', linewidth=2)
39 axes[1, 1].set_title('Second-Order: $\\Phi(\\lambda)$')
40
41 for ax in axes.flat:
42     ax.set_xlabel('$\\lambda$' if ax in axes[1] else '$\\Phi$')
43 plt.tight_layout()
44 plt.savefig('figures/fig6_first_vs_second.pdf', dpi=300)

```

Listing 6: First-order versus second-order transition comparison (Figure 5).

A.7 Figure Generation: Bimodal Distribution

```

1 import numpy as np
2 import matplotlib.pyplot as plt
3
4 np.random.seed(42)
5
6 def sample_phi(lam, n_samples=5000, lam_c=0.5):
7     """Sample Phi from a double-well potential."""
8     if lam < lam_c - 0.15:
9         return np.random.normal(0.3, 0.08, n_samples)
10    elif lam > lam_c + 0.15:
11        return np.random.normal(2.2, 0.1, n_samples)
12    else:
13        # Bimodal: mixture
14        p_high = (lam - (lam_c - 0.15)) / 0.3
15        n_high = int(p_high * n_samples)
16        n_low = n_samples - n_high
17        low = np.random.normal(0.3, 0.1, n_low)
18        high = np.random.normal(2.2, 0.12, n_high)
19        return np.concatenate([low, high])
20
21 lambdas = [0.2, 0.4, 0.5, 0.6, 0.8]
22 colors = ['#1b9e77', '#d95f02', '#7570b3', '#e7298a', '#66a61e']
23
24 fig, axes = plt.subplots(1, 5, figsize=(14, 3), sharey=True)

```

```

25 for ax, lam, color in zip(axes, lambdas, colors):
26     samples = sample_phi(lam)
27     ax.hist(samples, bins=40, density=True, color=color, alpha=0.7,
28             edgecolor='white', linewidth=0.5)
29     ax.set_title(f'$\\lambda = {lam}$', fontsize=11)
30     ax.set_xlabel('$\\Phi$', fontsize=10)
31     ax.set_xlim(-0.5, 3.5)
32 axes[0].set_ylabel('Density', fontsize=10)
33 plt.suptitle('Distribution of $\\Phi$ at Different Control '
34             'Parameter Values', fontsize=12, y=1.02)
35 plt.tight_layout()
36 plt.savefig('figures/fig7_bimodal.pdf', dpi=300, bbox_inches='tight')

```

Listing 7: Bimodal distribution of Φ near the transition (Figure 6).

Peer Reviews and Revision History

This section documents the peer review process and subsequent revisions, preserving the scholarly record of how the paper evolved in response to critical feedback.

First Round

Reviewer Recommendation: Major Revisions Required

This paper attempts an ambitious synthesis of three threads: a Bayesian argument about the evidential limits of consciousness science, a dynamical systems framework for phase transitions in integration measures, and a methodological proposal for separating levels of scientific claim. The writing is clear and well-structured, and the self-critical apparatus is a genuinely unusual and commendable feature. However, I have significant concerns about the novelty of the epistemological argument, the relationship between the toy models and the theoretical claims, and several gaps in the engagement with existing literature.

Issue 1: Novelty of the Bayesian Argument

Reviewer

The core concern: this result is not new. Theorem 2.3 is the zombie thought experiment restated in Bayesian notation. The paper would benefit from engagement with existing literature: Chalmers (1996) discusses the epistemic consequences at length; the knowledge argument (Jackson, 1982) is not cited; Nida-Rümelin (2023) and Stoljar (2006) address precisely this question. The trilemma should acknowledge variants discussed in the Type-A/Type-B physicalism debate (Chalmers, 2002).

Author Response

We accept this criticism. In the revised manuscript we have added a subsubsection “Relation to Existing Epistemic Arguments” that explicitly situates our result relative to Jackson (1982), Levine (1983), Stoljar (2006), Nida-Rümelin (2023), and Chalmers (2002). We now clearly state that our contribution is not the observation that functional and phenomenal descriptions come apart, but the formalisation within Bayesian epistemology that makes consequences for scientific methodology precise: the likelihood ratio is identically one for every empirical methodology in consciousness science.

Issue 2: Treatment of IIT’s Ontological Commitments

Reviewer

The paper treats “functional” as unambiguous, but IIT denies that Φ is merely functional. On IIT’s view, the causal structure *is* the experience. The paper should engage with Tononi et al.’s (2015) arguments for why the identity claim is constitutive of the formalism, not a detachable Level 3 addition.

Author Response

We have added “Engagement with IIT’s Ontological Commitments” acknowledging that IIT’s identity claim is constitutively linked to the formalism. We engage with Tononi & Koch (2015) and respond: even constitutive identities must be evaluated against alternatives, and no functional observation can adjudicate. We reference adversarial collaborations (Melloni et al., 2021) and comparative reviews (Seth & Bayne, 2022) as evidence that agnosticism about the identity claim is increasingly common.

Issue 3: Toy Models Described as Computational Confirmations**Reviewer**

The fundamental problem: the toy models do not compute what they claim. The “computational results” are analytical curves with added noise, not genuine simulations. Figure 3 computes τ from an analytical formula; Figure 4 uses a parametric sigmoid, not stochastic block model networks; Figure 7 samples from hardcoded Gaussian mixtures, not a Boltzmann machine. The paper should either perform genuine simulations or clearly label these as schematic illustrations.

Author Response

This is the most substantive criticism, and we accept it without qualification. Section 5 has been retitled “Schematic Illustrations and Analytical Models.” A new opening subsection explicitly states that the figures are generated from the analytical framework, not from network simulations. All figure captions have been rewritten to accurately describe their generating method. We cite Toker et al. (2019) on computational intractability and Mediano et al. (2022) on proxy measures, and identify genuine simulation as an important direction for future work. We consider this correction essential to the paper’s integrity.

Issue 4: Proposition 4.10 Without Proof**Reviewer**

The finite-size analysis (Proposition 4.10) is presented without proof. The effective spatial dimension d is not well-defined for cortical systems. The numerical estimates should be substantiated or flagged as speculative.

Author Response

A proof sketch has been added for all three parts of Proposition 4.10 based on standard finite-size scaling theory. A caveat paragraph addresses the ambiguity of the effective spatial dimension, referencing neural criticality literature (Beggs & Plenz, 2003; Cocchi et al., 2017). Numerical estimates are now presented as order-of-magnitude guides.

Issue 5: Literature Engagement

Reviewer

The reference list is notably thin (14 references). Significant omissions: Global Neuronal Workspace empirical literature, Seth & Bayne (2022), adversarial collaborations, Mediano et al. (2022), criticality literature, and the philosophical epistemic gap literature.

Author Response

The revised manuscript adds 15 references integrated at appropriate points: Jackson (1982), Levine (1983), Stoljar (2006), Nida-Rümelin (2023), Chalmers (2002) in Section 2; Seth & Bayne (2022), Melloni et al. (2021), Tononi & Koch (2015) in Sections 3–4; Beggs & Plenz (2003), Cocchi et al. (2017) in Section 4; Mediano et al. (2022), Toker et al. (2019) in Sections 4–5; Dehaene & Changeux (2011), Mashour et al. (2020) in Section 4. The reference list has expanded from 14 to 29 entries.

Issue 6: Structural Concerns

Reviewer

The Self-Critique boxes become repetitive. Table 4 classifying theories by transition type is speculative. The paper's length is excessive for the content delivered.

Author Response

We retain the Self-Critique boxes as a deliberate methodological commitment to transparent self-assessment but acknowledge the reviewer's point. Table 4 remains as a summary with the caveat that detailed dynamical analysis of each theory is beyond the present scope. We have not reduced the paper's length at this stage, as detailed proofs and explanations serve the stated goal of accessibility across disciplines. We are open to condensation if the editor considers it necessary.

Revision Log

Second Round: *Pending.*

Third Round: *Pending.*